

# Cambridge IGCSE Economics 0455

## 学生教材 · 逐课深度双语版

细粒度知识讲解 · 中英双语 · 真题化应用 · 图表推演

依据 Cambridge IGCSE Economics 0455

课程内容、公开样卷与评分要求编写。题型示例、情境、数字和图表均为原创改编，用于说明常见考点，不复制历年试题。

资料口径: Cambridge International 0455 syllabus and specimen assessment materials, accessed September 2026.

# 第 1 课 | 2.1 稀缺、选择与机会成本 ( Scarcity, choice and opportunity cost )

所属大章: 基本经济问题 | The basic economic problem

## 知识点精讲 | Core knowledge

稀缺 ( scarcity ) 不是 “东西很少” , 而是资源不足以同时满足所有欲望。空气通常不稀缺, 但在潜水瓶、医院氧气或严重污染情境中会变成稀缺资源。稀缺迫使家庭、企业、政府作出选择 ( choice ) 。每一次选择的真正代价是机会成本 ( opportunity cost ) : 被放弃的 “最佳替代方案” 的价值, 而不是所有放弃物的总和。

例子: 周六下午你可做兼职赚 120 元, 或复习经济学。若选择复习, 机会成本是 120 元, 而非 “120 元加上休息时间” 。答题时写: The opportunity cost of studying is the income forgone from the best alternative use of time.

## 深度展开与真题化应用 | In-depth development

### 概念边界与生活化理解

稀缺必须同时包含两个条件: 人的欲望很多, 而可用于满足欲望的资源有限。因此, 价格很高不等于稀缺, 免费物品也不一定永远不稀缺。学校只有一间实验室, 政府只有一笔预算, 家庭一天只有 24 小时, 三者都面对资源约束。选择发生后才出现机会成本, 而且机会成本只计算 next best alternative forgone ( 被放弃的最佳替代方案 ) 。

### 从选择推到经济后果

假设市政府可用 2

亿元修建医院或体育馆。若医院是最佳替代用途, 选择体育馆的机会成本就是医院带来的医疗服务, 而不是 “2 亿元” 本身, 因为这笔钱在两种方案中都会使用。选择还可能改变不同群体的福利: 体育馆使用者受益, 等待医疗的人可能受损。IGCSE 题目中的高分答案不仅写定义, 还要把有限资源、互斥用途和最佳替代方案连接起来。

### English development

Scarcity is a relative concept: resources are limited in relation to unlimited wants. It exists in rich as well as poor economies. Choice is unavoidable, and the opportunity cost of a decision is the benefit from the next best alternative that is given up. Money expenditure and opportunity cost are not always the same. A government project may cost \$10 million, but its economic opportunity cost is the best public service that could have been provided with those resources.

### 真题化知识应用

题型示例: A student can attend a revision course, work for \$45, or play football. The course is chosen and working is the next preferred option. What is the opportunity cost? 正确思路是 \$45 的收入, 而不是工作与足球的总和。结构题若问政府增加国防支出的 opportunity cost, 应回答可能少提供的教育、医疗或基础设施, 并结合题干判断哪一个最佳替代用途。

## 逐题精讲与评价拓展 | Exam workshop

### 进一步拆解: 这一课真正考什么

把 scarcity、choice 与 opportunity cost 看成一条连续逻辑: 资源有限并不会自动给出答案; 决策者必须比较互斥用途, 选择其中一个, 才会放弃最佳替代方案。家庭、企业和政府面对的资源不同, 但分析方法完全相同。

### 完整因果链 | Cause-and-effect chain

政府税收有限 → 同一批建筑工人和材料不能同时建设医院与机场 → 选择机场 → 医院服务成为 opportunity cost → 不同居民群体受到不同影响。写分析题时把每个箭头改成 because 或 therefore 连接的完整句子；如果中间一步无法解释，说明链条仍有跳跃。

### 数字与情境示范 | Numerical or contextual example

某学生有 3 小时，可复习、兼职赚 \$36 或参加价值最高为 \$20 的活动。选择复习且兼职是次优方案时，机会成本是 \$36，不是 \$56，也不是 3 小时。时间是稀缺资源，\$36 是最佳替代用途的收益。

### English guided explanation

Scarcity creates a constraint, choice is the decision made under that constraint, and opportunity cost measures the value of the next best option forgone. The alternative must be feasible and mutually exclusive with the chosen option. A complete answer identifies both the decision-maker and the specific benefit sacrificed.

### Cambridge-style multiple-choice application

Question. A council can build a library, repair a road or leave the money in a bank. It builds the library; road repair was its second preference. What is the opportunity cost?

Answer and reasoning. The benefit from repairing the road. The bank deposit is another alternative but is not the next best one. 选择题不仅要找到正确项，还要说明其余方向为什么与定义、公式或因果关系不符。

### Cambridge-style structured application

Question. Explain why economic growth does not remove scarcity.

Model development. Growth shifts productive capacity outwards and allows more wants to be satisfied. However, human wants can continue to expand and resources remain limited relative to those wants. The economy therefore still has to choose how to use labour, land and capital, so opportunity cost remains.

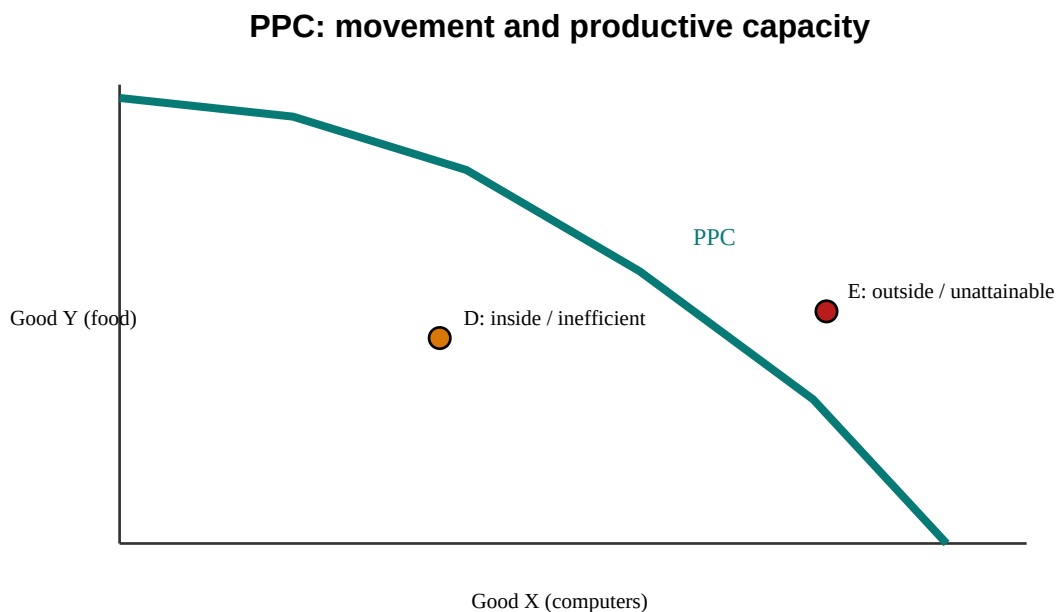
### 高频混淆点 | Common misconception

不要把 shortage 与 scarcity 混用。Shortage 是某价格下  $Q_d > Q_s$ ，价格变化可消除；scarcity 是资源相对于欲望有限，是所有经济体长期面对的基本问题。

### 条件与评价 | Conditions and evaluation

机会成本的大小取决于决策者偏好和可用信息。同一选择对学生、企业或政府可能有不同的 next best alternative。评价不是另起一个无关观点，而是改变时间、幅度、弹性、资源条件或利益相关者后，检查原结论是否仍成立。

## 图表推演 | Diagram analysis



考试图表必须标出坐标轴、曲线名称、新旧均衡和变化方向。文字解释需说明原因、曲线移动以及价格或数量结果。

## 英文精讲 | English explanation

Scarcity exists because resources are finite while wants are unlimited. It forces choice, and every choice has an opportunity cost: the next best alternative forgone.

## IGCSE 大考口径 | Cambridge exam focus

大考口径：准确定义；紧扣题干应用；解释完整因果链；评价结论说明条件。

## 第 2 课 | 2.2 四种生产要素与报酬 ( Factors of production )

所属大章: 基本经济问题 | The basic economic problem

### 知识点精讲 | Core knowledge

Land ( 土地 ): 自然资源, 如耕地、矿产、森林; 报酬为 rent ( 地租 )。

Labour ( 劳动 ): 人提供的体力与脑力; 报酬为 wages ( 工资 )。

Capital ( 资本 ): 用于生产的人造物品, 如机器、厂房、软件; 报酬为 interest ( 利息 )。它不是“有钱”本身。

Enterprise ( 企业家才能 ): 组织资源、承担风险、创新和作决策; 报酬为 profit ( 利润 )。

常见误区: CEO 的工作属于 labour; 一辆消费者自用汽车不是 capital, 但出租车公司用于提供运输服务的汽车是 capital。

### 深度展开与真题化应用 | In-depth development

#### 四种要素不是四种“东西”

Land 包括自然提供的资源, 不只是一块土地; 海洋、太阳能、矿藏都属于 land。Labour 是人的体力和脑力投入, 数量之外还要看技能、健康和教育。Capital 是为生产其他产品而制造的生产资料, 银行账户里的钱是 finance, 不自动等于 capital。Enterprise 负责组织其他要素、承担不确定性并作出创新决策。同一个物品的分类取决于用途: 家庭自用电脑是消费品, 设计公司用于工作的电脑是 capital good。

#### 数量、质量、流动性与报酬

生产能力不仅由要素数量决定, 还取决于质量和 mobility。培训提升 labour quality; 新机器提高 capital quality; 灌溉可提高 land productivity。工人可能因住房、资格或语言限制而不能从衰退行业转到扩张行业, 这叫 occupational/geographical immobility。要素报酬通常写作 land-rent、labour-wages、capital-interest、enterprise-profit, 但现实中企业家也可能领取工资, 考试应按其所承担的经济功能分类。

#### English development

Factors of production are resources used to produce goods and services. Labour productivity is output per worker or per hour, not total output. Investment adds to the stock of capital goods, while depreciation reduces it. Enterprise is distinct because the entrepreneur combines resources, takes risks and may introduce new products or production methods.

#### 真题化知识应用

选择题常给出 drilling machinery、delivery driver、forest 和 business owner, 要求分别识别 capital、labour、land 和 enterprise。问答题“Explain two causes of higher labour productivity”需要两条展开链: education raises skills, so each worker can produce more per hour; better machinery allows workers to complete tasks faster, raising output per worker。

### 逐题精讲与评价拓展 | Exam workshop

#### 进一步拆解: 这一课真正考什么

四类要素要按经济功能分类, 而不是按外观分类。自然形成的是 land; 人的服务是 labour; 用于生产的制造品是 capital; 组织、创新和承担风险是 enterprise。要进一步区分要素数量、质量、生产率和流动性。

#### 完整因果链 | Cause-and-effect chain

职业培训 → worker skills improve → output per hour rises → unit labour cost may fall → firm becomes more competitive → demand for skilled labour and wages may rise。写分析题时把每个箭头改成 because 或 therefore 连接的完整句子；如果中间一步无法解释，说明链条仍有跳跃。

### 数字与情境示范 | Numerical or contextual example

五名工人每天生产 200 件，labour productivity=200/5=40 件/人。培训后四名工人生产 200 件，生产率=50 件/人，提高 25%。总产量没变，但单位劳动投入效率提高。

### English guided explanation

The classification of a resource depends on how it is used. A van owned by a delivery firm is capital, while the driver provides labour. Productivity is output per unit of input. An increase in the number of workers can raise total output without raising labour productivity.

### Cambridge-style multiple-choice application

Question. Which change definitely improves the quality of labour: more mineral deposits, worker training, a larger factory, or a lower interest rate?

Answer and reasoning. Worker training, because it increases human capital and the skills embodied in labour. 选择题不仅要找到正确项，还要说明其余方向为什么与定义、公式或因果关系不符。

### Cambridge-style structured application

Question. Analyse how an increase in the quantity and quality of capital may affect an economy.

Model development. More and better machinery can raise output per worker. Higher productivity may lower average cost, enabling firms to reduce prices or increase profit. Competitiveness and exports may rise, increasing real GDP. The effect depends on workers having the skills to operate the capital and on sufficient demand.

### 高频混淆点 | Common misconception

Money used to buy a machine is finance; the machine itself is capital. Shares and banknotes are financial assets, not automatically factors of production.

### 条件与评价 | Conditions and evaluation

要素增加的效果取决于互补要素。例如只有机器而没有电力、技能或维修，productive capacity 未必充分转化为 actual output。

评价不是另起一个无关观点，而是改变时间、幅度、弹性、资源条件或利益相关者后，检查原结论是否仍成立。

## 英文精讲 | English explanation

Land, labour, capital and enterprise are inputs used in production. Their rewards are rent, wages, interest and profit.

## IGCSE 大考口径 | Cambridge exam focus

大考口径：准确定义；紧扣题干应用；解释完整因果链；评价结论说明条件。

## 第 3 课 | 2.3 生产可能性曲线 ( Production possibility curve, PPC )

所属篇章：基本经济问题 | The basic economic problem

### 知识点精讲 | Core knowledge

PPC 显示在既定资源与技术下，一个经济体能够生产两种产品的最大组合。曲线上任一点是 productively efficient (生产有效率)；曲线内表示资源闲置或失业；曲线外短期不可达到。曲线向外移动来自更多/更好的资源、教育培训、技术进步和投资；战争、自然灾害或劳动力流失会使曲线向内移动。

若 PPC 向原点凹，表示生产一种产品越多，转移资源的适配度越差，opportunity cost increases (机会成本递增)。画图题务必写坐标、两种产品、曲线与移动箭头；不要把“沿 PPC 移动”(产品组合改变)误画成“PPC 整条移动”(生产能力改变)。

#### PPC 图像的读法：从坐标到结论

先画两条互相垂直的轴：横轴写 Good X (例如 computers)，纵轴写 Good Y (例如 food)，并标出原点 O。曲线连接两个截距：若所有资源生产食物，位于纵轴截距；若所有资源生产电脑，位于横轴截距。曲线上的 A、B、C 是不同的可行组合，不是三个“效率等级”。从 A 沿曲线向 B 移动，表示资源从食物部门转向电脑部门；生产能力没有改变，只是组合改变。

图中曲线以内的点 D 代表 productive inefficiency (生产无效率)：在不减少另一种产品的情况下仍可增加产出，原因可能是失业、机器闲置或管理低效。曲线以外的点 E 在当前资源和技术下不可达。若培训提高劳动生产率，整条 PPC 向外移；若地震摧毁厂房，整条 PPC 向内移。答题时用箭头标示 shift，并明确写出“productive potential changes”。

#### 为什么曲线通常向原点凹？机会成本的推演

假设经济体只有土地、劳动和资本，生产电脑和食物。最初从食物转出少量资源时，可以先转移最适合电脑生产的工人和设备，所以电脑增加 1 单位只牺牲 1 单位食物。继续转移时，电脑部门需要使用原本更适合种植的土地和农业工人，牺牲的食物增加到 2、3、4 单位。这就是 increasing opportunity cost (机会成本递增)，图上表现为越接近横轴，曲线越陡。

由表可见，每增加相同的 10 台电脑，放弃的食物越来越多。若题目要求“calculate opportunity cost”，要写  $\text{opportunity cost} = \text{output forgone} / \text{output gained}$ ，并说明是哪个区间；不能只写“机会成本为 14”，因为不同区间的数值不同。

#### 课堂例题：区分 movement 与 shift

某国从组合 A (20 computers, 88 tonnes of food) 变为组合 B (30 computers, 74 tonnes of food)。第一问：发生了什么？答案是沿 PPC 移动，电脑增加 10，食物减少 14；机会成本为 14 吨食物。第二问：若新工厂和自动化使每一种产品的最大产量都提高，如何作图？答案是 PPC outward shift；原曲线上的 A、B 仍是旧生产能力下的组合，新曲线应位于右上方。

高分英语表达：The economy moves from A to B along the same PPC, so resources are reallocated and the opportunity cost of 10 additional computers is 14 tonnes of food. In contrast, an outward shift would indicate an increase in productive potential caused by better resources or technology.

#### 画图与评价的评分清单

1. 坐标轴有名称和方向；2. 曲线标为 PPC/PPC；3. 点标 A、B、inside、outside；4. 移动用沿曲线箭头，能力变化用整条曲线箭头；5. 解释原因，而非只写“向外”；6. 写出短期可行性和长期增长的区别。若资源只适合其中一种产品，PPC 可接近直线；若资源完全不能互换，机会成本近似不变。



## 深度展开与真题化应用 | In-depth development

### PPC 是约束图，不是需求供给图

PPC 的每一个点都对应两种产品的产量组合，并假定资源数量、资源质量和技术暂时不变。曲线上点表示所有现有资源已被充分有效利用；曲线内点表示实际产量低于潜在产量，原因可能是 cyclical unemployment、机器闲置或组织低效；曲线外点不是“永远不可能”，而是以当前生产能力不能持续达到。贸易可以让消费点位于本国 PPC 外，但并不等于本国生产点在曲线外。

### 机会成本、曲率与偏向性移动

从 A(0 computers, 100 food) 到 B(10, 96)，10 台电脑的机会成本是 4 单位 food；再到 C(20, 88)，新增 10 台电脑的机会成本变为 8。机会成本递增的原因不是曲线“天生弯”，而是生产要素具有 specialised uses。若只有电脑技术进步，PPC

主要绕食物轴截距向外旋转；若教育普遍提高生产率，两端都可能外移。沿曲线移动表示 resource reallocation，不表示 growth。

### English development

A production possibility curve shows the maximum attainable combinations of two products when resources and technology are fixed. Productive efficiency concerns whether output is maximised; allocative efficiency concerns whether the chosen mix matches society's preferences. An outward shift indicates an increase in productive potential, but actual output will rise only if the additional capacity is used.

### 真题化知识应用

常见图题给出 recession、training、natural disaster 或资源从消费品转向资本品。Recession 通常使经济从曲线上点移到曲线内；training 使 PPC 外移；灾害使其内移；更多 capital goods 会降低当前 consumer goods，却可能使未来 PPC 外移更快。答图时依次写 axis labels、PPC1/PPC2、arrow 和原因。

## 逐题精讲与评价拓展 | Exam workshop

### 进一步拆解：这一课真正考什么

PPC 同时表达 scarcity、choice、opportunity cost、productive efficiency 和 economic growth。读图先看轴与点，再看是否沿曲线移动，最后看曲线是否整体移动。消费点可因贸易位于 PPC 外，但本国生产点不能长期位于当前 PPC 外。

### 完整因果链 | Cause-and-effect chain

更多资本品生产 → 当前消费品机会成本上升 → capital stock 随时间增加 → future productivity rises → PPC later shifts outwards → future consumption possibilities expand。写分析题时把每个箭头改成 because 或 therefore 连接的完整句子；如果中间一步无法解释，说明链条仍有跳跃。

### 数字与情境示范 | Numerical or contextual example

组合 A 为 0 台机器和 100 吨粮食，B 为 10 和 94，C 为 20 和 84。A 到 B 的 10 台机器成本 6 吨粮，B 到 C 成本 10 吨，显示 increasing opportunity cost。

### English guided explanation

A bowed-out PPC reflects specialised resources. As production of one good expands, resources increasingly unsuitable for that activity are transferred, so more of the other good must be sacrificed. A straight line indicates constant opportunity cost.

### Cambridge-style multiple-choice application



Question. An economy moves from a point inside its PPC to a point on the curve. What is the most likely explanation?

Answer and reasoning. Unemployment or inefficiency has fallen. Existing productive capacity is used more fully; the PPC itself need not shift.

选择题不仅要找到正确项，还要说明其余方向为什么与定义、公式或因果关系不符。

### Cambridge-style structured application

Question. Explain how investment in capital goods can affect current and future living standards.

Model development. Allocating more resources to capital goods requires less current production of consumer goods, so present consumption may fall. The larger capital stock can raise productivity and productive potential later, shifting the PPC outwards. Future real income and consumption may then increase, provided the capital is useful and fully employed.

### 高频混淆点 | Common misconception

曲线上的所有点都是 productively efficient, 但只有符合社会偏好的组合才是 allocatively efficient。两者不能互换。

### 条件与评价 | Conditions and evaluation

PPC 外移不保证生活水平同比提高。人口增长、污染、分配不均以及新增能力是否实际使用都会改变最终结果。评价不是另起一个无关观点，而是改变时间、幅度、弹性、资源条件或利益相关者后，检查原结论是否仍成立。

## 图表推演 | Diagram analysis

**PPC: movement and productive capacity**



考试图表必须标出坐标轴、曲线名称、新旧均衡和变化方向。文字解释需说明原因、曲线移动以及价格或数量结果。

## 英文精讲 | English explanation

A PPC shows the maximum combinations of two goods that can be produced with current resources and technology. Movement along changes the output mix; a shift changes productive potential.

## IGCSE 大考口径 | Cambridge exam focus

大考口径：准确定义；紧扣题干应用；解释完整因果链；评价结论说明条件。

## 第 4 课 | 2.4 经济制度与资源配置 (Economic systems)

所属大章: 基本经济问题 | The basic economic problem

### 知识点精讲 | Core knowledge

Market economy 主要由价格和利润信号配置资源; planned economy

由政府决定主要生产什么、如何生产、为谁生产; 现实多是 mixed economy (混合经济)。评价制度不能只列优缺点: 市场激励创新但可能出现不平等、公共品不足和外部性; 政府可纠正失灵但也可能信息不足、官僚低效。

### 深度展开与真题化应用 | In-depth development

#### 三种制度回答同样的三个问题

所有经济体都必须决定 what to produce, how to produce and for whom to produce。市场经济主要依靠价格、利润和消费者选择; 计划经济主要依靠政府指令和公共所有; 混合经济同时使用市场与政府。判断制度时不要看“有没有政府”或“有没有企业”, 而要看关键资源配置决定由谁作出。

#### 比较激励、信息、效率与公平

价格上涨向企业发出 scarcity 和 profitability

信号, 可能吸引资源进入该市场; 竞争鼓励降成本和创新。然而市场可能少供 public goods、忽略 external costs, 并依据购买力而非需要分配产品。政府可以提供医疗教育、征税或监管, 但也可能缺乏地方信息、反应缓慢或受到政治目标影响。现实结论通常取决于市场竞争程度、政府能力和产品性质。

#### English development

In a market economy, consumer demand and producer supply interact through the price mechanism. In a planned economy, the state owns or directs many resources. A mixed economy combines private decision-making with government intervention. Evaluation should compare outcomes, not merely list features: efficiency may improve with competition, while equity or public-good provision may require government action.

#### 真题化知识应用

若题目问 “Discuss whether resources should be allocated by the market”, 支持面可写 price signals、choice、profit incentive、competition; 反面要发展 market failure、inequality 与 information problems。结论不能写 “一半一半”, 应说明例如公共交通具有外部收益且竞争有限, 因此适度补贴和监管可能优于完全自由市场。

### 逐题精讲与评价拓展 | Exam workshop

#### 进一步拆解: 这一课真正考什么

比较制度时围绕 ownership、decision-making、incentives、consumer choice、equity 和 market failure, 不要只写 “政府多或少”。现实 mixed economy 中, 哪些行业由市场配置、哪些由国家提供, 比制度标签更重要。

#### 完整因果链 | Cause-and-effect chain

消费者对太阳能设备需求增加 → price/profit signal strengthens → firms enter and hire resources → supply expands → resources shift away from less profitable uses。计划经济则可能由政府根据能源目标直接配置资源。写分析题时把每个箭头改成 because 或 therefore 连接的完整句子; 如果中间一步无法解释, 说明链条仍有跳跃。

#### 数字与情境示范 | Numerical or contextual example

假设市场价格为 \$20 时有 1 000 名消费者愿买，而政府计划价 \$10 时需求量 1 500、供给量 700。低价提高可负担性，却产生 800 的 shortage，需要排队或行政配给。

### English guided explanation

A market system can process dispersed information through prices, while planning can direct resources towards social priorities. Neither system guarantees efficiency or fairness. A mixed economy attempts to combine market incentives with public provision and regulation.

### Cambridge-style multiple-choice application

Question. Which feature is most characteristic of a market economy: consumer sovereignty, state production targets, equal income, or free public goods?

Answer and reasoning. Consumer sovereignty, because purchases influence what profit-seeking firms produce. 选择题不仅要找到正确项，还要说明其余方向为什么与定义、公式或因果关系不符。

### Cambridge-style structured application

Question. Discuss whether healthcare should be allocated entirely by the market.

Model development. Market provision may encourage efficiency, choice and innovation. Competition can reduce cost. However, low-income patients may be unable to pay, consumers may lack medical information and healthcare creates external benefits. Public funding or regulation may improve access, although taxation has opportunity cost and government provision may be inefficient.

### 高频混淆点 | Common misconception

Market economy 不等于没有政府；planned economy 也不等于所有产品免费。价格、所有权和决策权才是关键。

### 条件与评价 | Conditions and evaluation

最佳组合取决于产品性质与政府能力。公共品、外部性强或信息不对称严重的领域通常需要更多干预，普通消费品更适合竞争。

评价不是另起一个无关观点，而是改变时间、幅度、弹性、资源条件或利益相关者后，检查原结论是否仍成立。

### 英文精讲 | English explanation

Market, planned and mixed economies differ in how resources are allocated. Compare incentives, information, efficiency and equity.

### IGCSE 大考口径 | Cambridge exam focus

大考口径：准确定义；紧扣题干应用；解释完整因果链；评价结论说明条件。

## 第 5 课 | 2.5 逐步判断：一个 PPC 题到底在问什么？

所属大章：基本经济问题 | The basic economic problem

### 知识点精讲 | Core knowledge

Step 1 – 先识别位置。点在曲线内：不是“没有稀缺”，而是既定资源没有充分利用；可用 unemployment、inefficient organisation 或 recession 解释。点在曲线上：在当前技术下生产有效率，但不必然符合社会偏好。点在曲线外：当前不可行，除非贸易、技术或资源改变。

Step 2 – 再识别变化类型。“More workers receive training” 提高 labour productivity, PPC outward shift; “workers move from food to computers” 是 movement along PPC; “an earthquake destroys factories” 是 inward shift。答题句式：An improvement in human capital raises the productive potential of the economy, so the PPC shifts outwards.

Step 3 – 最后说代价。若由 A 转向 B，机会成本是少生产的 A。请避免写“B 的机会成本是 B”；机会成本必须是放弃的另一个选项。

### 深度展开与真题化应用 | In-depth development

#### 先识别题目在改变什么

PPC 题只有三类核心变化：点的位置、沿曲线移动、整条曲线移动。失业下降使生产点由曲线内向曲线靠近；军用品增加且消费品减少是沿同一曲线移动；技术、劳动力、资本存量或自然资源改变才移动整条曲线。题干出现“产量上升”并不自动说明 PPC 外移，因为利用闲置资源也能提高实际产量。

#### 用反事实检查答案

问自己：资源和技术是否改变？若没有，就不能画新 PPC。再问：两种产品是否一增一减？若是，多半是沿曲线移动。若两种最大产量都提高，才通常整体外移。最后区分 short-run actual output 与 long-run productive potential。

#### English development

Use precise verbs: an economy moves along a PPC when it reallocates resources; the PPC shifts when productive capacity changes. Moving from an inefficient point towards the frontier is economic recovery, not necessarily long-run growth. A point on the frontier is productively efficient but may not be allocatively efficient.

#### 真题化知识应用

选择题干：“A country reduces unemployment without changing its labour force or technology.”

正确图形是从 PPC 内部移向曲线，而不是曲线外移。结构题可写：Lower unemployment brings idle labour into production, increasing actual output. However, the maximum possible combinations are unchanged, so the PPC itself does not shift.

### 逐题精讲与评价拓展 | Exam workshop

#### 进一步拆解：这一课真正考什么

把题干中的动词翻译成图形语言：reallocate 表示沿曲线；recover from recession 表示从曲线内向边界；increase resources/productivity/technology 表示 PPC shift；consume through trade 可能在生产边界外。

#### 完整因果链 | Cause-and-effect chain

地震摧毁 factories → capital stock falls → maximum attainable output falls → PPC shifts inward。重建只把能力恢复到原水平，不一定代表相对灾前增长。写分析题时把每个箭头改成 because 或

therefore 连接的完整句子；如果中间一步无法解释，说明链条仍有跳跃。

### 数字与情境示范 | Numerical or contextual example

旧 PPC 最大生产 50 台机器或 100 吨粮食；技术只改善机器生产，使最大机器升至 80、粮食仍为 100。图形绕粮食截距向外旋转，而不是平行移动。

### English guided explanation

Always identify whether the question concerns actual output, productive potential or the composition of output. These correspond to movement from inside the frontier, a shift of the frontier, and movement along the frontier.

### Cambridge-style multiple-choice application

Question. What happens to the PPC when unemployed workers find jobs but the labour force and productivity are unchanged?

Answer and reasoning. The production point moves towards the existing PPC. Productive potential is unchanged. 选择题不仅要找到正确项，还要说明其余方向为什么与定义、公式或因果关系不符。

### Cambridge-style structured application

Question. Analyse the PPC effects of a government transferring resources from consumer goods to capital goods.

Model development. In the current period, the economy moves along its PPC towards more capital goods and fewer consumer goods. The sacrificed consumer goods are the opportunity cost. If capital formation raises future productivity, the future PPC may shift outwards, allowing more of both goods.

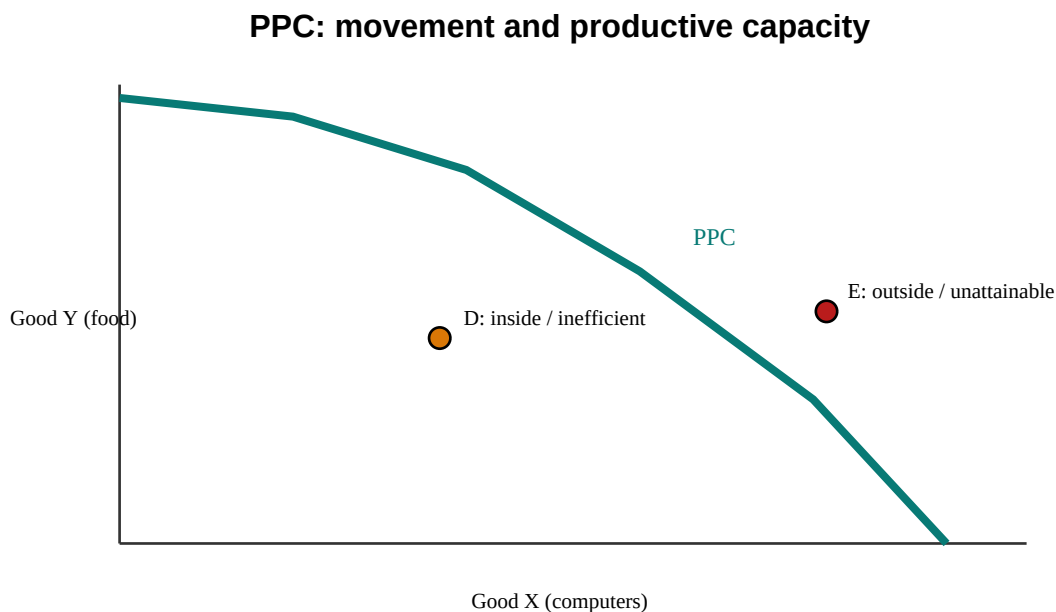
### 高频混淆点 | Common misconception

“Output rises” 不必然等于 “economic growth”。使用闲置资源提高 actual output 与 productive potential 增长必须分开。

### 条件与评价 | Conditions and evaluation

判断 outward shift 的幅度还要看投资效率、折旧、技能匹配和时间。不是所有资本支出都同样提高生产力。评价不是另起一个无关观点，而是改变时间、幅度、弹性、资源条件或利益相关者后，检查原结论是否仍成立。

## 图表推演 | Diagram analysis



考试图表必须标出坐标轴、曲线名称、新旧均衡和变化方向。文字解释需说明原因、曲线移动以及价格或数量结果。

## 英文精讲 | English explanation

A PPC shows the maximum combinations of two goods that can be produced with current resources and technology. Movement along changes the output mix; a shift changes productive potential.

## IGCSE 大考口径 | Cambridge exam focus

大考口径：准确定义；紧扣题干应用；解释完整因果链；评价结论说明条件。



## 第 6 课 | 2.1 市场、买方与卖方 | Market, buyers and sellers

所属大章: 资源配置 | The allocation of resources

### 知识点精讲 | Core knowledge

市场 (market) 是使买者与卖者相互接触、进行交换的一种安排; 它可以是实体市场, 也可以是网络平台。买方通常是 consumers, 他们购买商品与服务满足需要; 卖方通常是 firms, 他们组织生产要素并提供产品。市场不要求买卖双方在同一地点。

English explanation. A market is an arrangement that brings buyers and sellers into contact. Buyers express demand; sellers express supply. Their decisions interact through prices. In Cambridge questions, identify the product market first and distinguish a consumer from a producer.

### 深度展开与真题化应用 | In-depth development

#### 市场是一种安排

市场不一定有店铺或固定地点。网约车平台、外汇市场和劳动力市场都让潜在买者与卖者交换。买者形成 demand, 卖者形成 supply, 价格承担 rationing、signalling 和 incentive 功能。分析前要先界定市场: 讨论“饮料市场”与“无糖饮料市场”可能得到不同的竞争者和替代品。

#### 市场参与者的目标与约束

消费者通常希望在预算内提高 satisfaction; 企业希望实现利润或其他目标, 但受成本、技术和竞争限制。政府也可能作为买方采购疫苗, 或作为卖方提供交通服务。市场中的“买方”和“卖方”是经济角色, 不是固定身份; 家庭在产品市场是买方, 在劳动力市场却出售 labour services。

#### English development

A market is any arrangement that brings buyers and sellers into contact. It includes all potential as well as actual buyers and sellers. The price provides information about relative scarcity and helps coordinate independent decisions. Always identify the specific product, geographical area and time period when interpreting market data.

#### 真题化知识应用

选择题会用 online auction 迷惑学生: 没有实体地点仍是 market。结构题“Explain the role of prices”可写: A higher price signals stronger demand or greater scarcity; it rations the product to consumers willing and able to pay; it gives producers an incentive to increase supply because potential revenue rises。

### 逐题精讲与评价拓展 | Exam workshop

#### 进一步拆解: 这一课真正考什么

市场的范围由 product、place、buyers/sellers 和 time 界定。家庭、企业与政府都可能转换角色: 家庭在商品市场买产品, 在劳动市场卖劳动; 企业买生产要素并卖产品。价格是联系这些决定的共同变量。

#### 完整因果链 | Cause-and-effect chain

消费者评价产品并形成 demand → sellers compare price with cost → transactions reveal sales and stock information → prices adjust → resources move between markets。写分析题时把每个箭头改成 because 或 therefore 连接的完整句子; 如果中间一步无法解释, 说明链条仍有跳跃。

#### 数字与情境示范 | Numerical or contextual example

同一饮料在学校商店售价 \$2、线上送货价

\$3。若送货服务属于产品的一部分，这可能是两个相关但不完全相同的市场，不能直接认定存在无风险套利。

### English guided explanation

A market includes potential transactions, not only completed sales. Its boundaries matter because a narrow definition may show few competitors, while a broad definition may include many substitutes. Online interaction is sufficient to bring buyers and sellers into contact.

### Cambridge-style multiple-choice application

Question. Which is a market: only a physical shop, only a stock exchange, any arrangement connecting buyers and sellers, or any government office?

Answer and reasoning. Any arrangement connecting buyers and sellers. Physical location is not required. 选择题不仅要找到正确项，还要说明其余方向为什么与定义、公式或因果关系不符。

### Cambridge-style structured application

Question. Explain two ways prices influence buyers and sellers in a market.

Model development. A higher price rations the product by reducing the quantity consumers are willing and able to buy. It also provides sellers with an incentive to increase quantity supplied because potential revenue and profit rise. These responses help coordinate the two sides of the market.

### 高频混淆点 | Common misconception

Consumer 是为了自身或家庭使用而购买者；购买原料的 firm 是 buyer，却不是最终 consumer。

### 条件与评价 | Conditions and evaluation

数字平台降低搜寻成本，但平台算法、网络效应和信息不对称也会影响竞争，市场不一定因此完全有效。评价不是另起一个无关观点，而是改变时间、幅度、弹性、资源条件或利益相关者后，检查原结论是否仍成立。

## IGCSE 大考口径 | Cambridge exam focus

大考口径：准确定义；紧扣题干应用；解释完整因果链；评价结论说明条件。

## 第 7 课 | 2.2 需求、个人需求与市场需求 | Individual and market demand

所属大章: 资源配置 | The allocation of resources

### 知识点精讲 | Core knowledge

需求 (demand) 是消费者在一定时期、不同价格下愿意并且能够购买的数量。个人需求是一个消费者的需求; 市场需求是在每一个价格水平把所有消费者的需求量横向相加。若价格为 \$10, A 购买 500 单位, B 购买 400 单位, 则市场需求量为 900。

English explanation. Demand is the willingness and ability to buy a product at different prices over a period of time. Market demand is the horizontal sum of all individual demand: quantities, not prices, are added at each price.

### 深度展开与真题化应用 | In-depth development

Demand 包含 willing、able、price 与 time

想买但付不起不是有效需求, 有能力但不愿买也不是需求。需求表列出一定时期内不同价格对应的 quantity demanded, 曲线只是把表格图形化。需求定律说明 ceteris paribus 下价格越低, 需求量通常越大, 因为替代效应和收入效应共同作用。

市场需求为何是横向加总

同一个价格下, 把每位消费者愿意购买的数量相加。例如 \$10 时 A 买 5 个、B 买 4 个, market quantity demanded 为 9 个; 不能把两个 \$10 相加为 \$20。若第三位消费者在该价位愿意买 2 个, 市场需求量变成 11, 整条市场需求曲线也可能向右。

English development

Demand is the willingness and ability to buy a product at different prices over a given period. Individual demand refers to one consumer. Market demand is obtained by adding the quantities demanded by all consumers at each price. The demand curve normally slopes downward because a lower price increases purchasing power and makes the product cheaper relative to substitutes.

真题化知识应用

表格题给出 price 8、6、4 与两位买者数量, 先逐行加总 quantity, 再作图。若问 “Explain why market demand for bus travel may rise”, 答案应使用人口增加、汽油价格上涨 (替代品) 或收入下降且公交为 inferior service 等非价格因素, 而不是写公交票价下降。

### 逐题精讲与评价拓展 | Exam workshop

进一步拆解: 这一课真正考什么

Demand 定义同时包括 willingness、ability、different prices 和 period of time。需求表是原始关系, 需求曲线是图形表达。Market demand 必须在每一个相同价格横向加总所有 individual quantities。

完整因果链 | Cause-and-effect chain

price falls ceteris paribus → product becomes cheaper relative to substitutes and purchasing power rises → quantity demanded increases → movement down the same demand curve。写分析题时把每个箭头改成 because 或 therefore 连接的完整句子; 如果中间一步无法解释, 说明链条仍有跳跃。

## 数字与情境示范 | Numerical or contextual example

价格 \$12 时 A 买 3、B 买 5、C 买 0，市场需求量 8；价格 \$8 时分别买 5、8、2，市场需求量 15。每行加数量，不能加价格。

## English guided explanation

Willingness without purchasing power is a desire, not effective demand. Ability without willingness also produces no demand. The time period matters: ten units per day and ten units per year describe very different demand.

## Cambridge-style multiple-choice application

Question. At a price of \$5, three consumers demand 4, 7 and 9 units. What is market quantity demanded?

Answer and reasoning. 20 units. Market demand is the horizontal sum of individual quantities at the same price. 选择题不仅要找到正确项，还要说明其余方向为什么与定义、公式或因果关系不符。

## Cambridge-style structured application

Question. Explain why a demand curve usually slopes downwards.

Model development. When price falls, the product becomes cheaper relative to substitutes, so consumers may switch towards it. The lower price also increases real purchasing power, enabling consumers to buy more. Hence quantity demanded rises, *ceteris paribus*.

## 高频混淆点 | Common misconception

Market demand 不是平均需求，也不是把 individual demand curves 的价格纵向相加。

## 条件与评价 | Conditions and evaluation

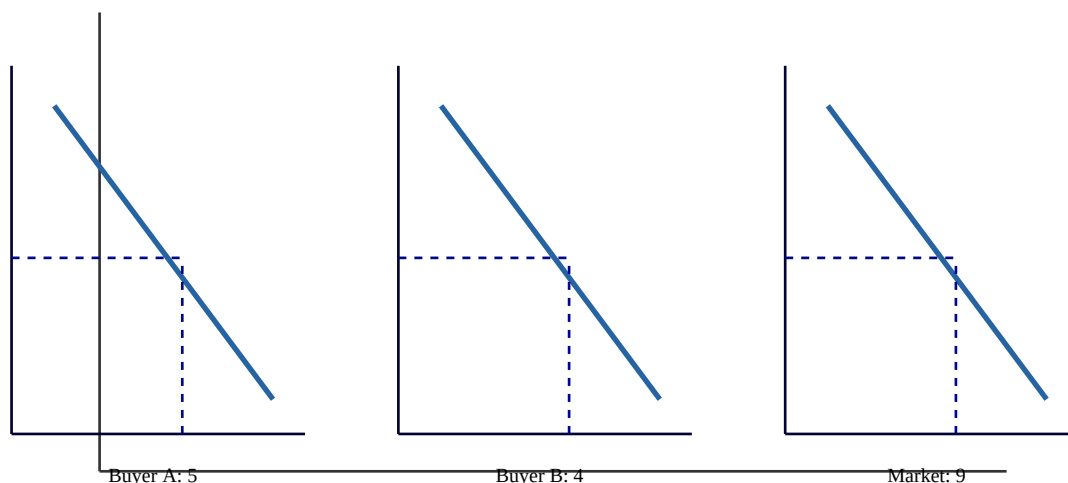
某些特殊情形可能不符合普通需求定律，但 0455 常规模型默认 *ceteris paribus*

和向下需求曲线，除非题干明确提供异常原因。

评价不是另起一个无关观点，而是改变时间、幅度、弹性、资源条件或利益相关者后，检查原结论是否仍成立。

## 图表推演 | Diagram analysis

### Market demand = sum of individual demand



考试图表必须标出坐标轴、曲线名称、新旧均衡和变化方向。文字解释需说明原因、曲线移动以及价格或数量结果。

## IGCSE 大考口径 | Cambridge exam focus

大考计算口径：在同一价格水平横向相加各消费者的 quantity demanded，不是把价格相加。

## 第 8 课 | 2.3 需求量移动与需求曲线移位 | Movement along and shift of demand

所属篇章：资源配置 | The allocation of resources

### 知识点精讲 | Core knowledge

商品自身价格下降会导致 extension in demand / increase in quantity demanded, 表现为沿同一条需求曲线向右下移动; 自身价格上升会导致 contraction。收入、相关商品价格、广告、偏好、人口结构、天气和未来价格预期属于非价格决定因素, 它们改变整条需求曲线。正常品收入增加使需求右移; 劣等品收入增加可能使需求左移。替代品价格上涨会增加本商品需求; 互补品价格上涨会减少本商品需求。

English explanation. A change in the product's own price causes a movement along the existing demand curve. A non-price determinant causes the whole curve to shift. Write "quantity demanded increases" for an extension and "demand increases" for a rightward shift.

### 深度展开与真题化应用 | In-depth development

#### 一条曲线还是两条曲线

判断的第一问永远是: 变的是商品自己的价格, 还是其他条件? Own price change 只改变 quantity demanded, 点沿原曲线移动; non-price determinant 改变 demand, 整条曲线移动。价格下降形成 extension in demand, 价格上涨形成 contraction in demand。收入、偏好、广告、人口、预期和相关商品价格不在纵轴上, 所以需要画新曲线。

#### 相关商品与收入的方向判断

咖啡与茶是 substitutes: 咖啡价格上升, 部分消费者改买茶, 茶的需求量 increases。汽车与汽油是 complements: 汽油价格上升使驾车总成本增加, 汽车需求下降。收入上升令 normal good 需求增加, 却可能令 inferior good 需求减少。先识别关系, 再写消费者选择改变, 最后判断  $D_0 \rightarrow D_1$  的方向。

#### English development

An extension of demand is an increase in quantity demanded caused only by a fall in the product's own price. An increase in demand is a rightward shift caused by a non-price determinant. These expressions are not interchangeable. In written answers, name the determinant, explain the change in willingness and ability to buy, and state the direction of the shift.

#### 真题化知识应用

仿真选择题: The price of smartphones is unchanged, but mobile-data charges fall. Smartphone demand is likely to increase because the services are complementary。图上应保持 smartphone price 轴不变, 把  $D_1$  向右移到  $D_2$ 。若手机自身降价, 则只能在  $D_1$  上由 A 移到 B。

### 逐题精讲与评价拓展 | Exam workshop

#### 进一步拆解: 这一课真正考什么

先圈出变化变量。Own price 在纵轴上, 变化造成 movement; income、taste、advertising、population、expectations 与 related goods prices 不在纵轴上, 变化造成 shift。语言上 extension/contraction 对应 quantity demanded, increase/decrease 对应 demand。

#### 完整因果链 | Cause-and-effect chain

tea price rises → coffee is a substitute → consumers switch from tea to coffee → coffee demand increases → coffee demand curve shifts right → coffee equilibrium price and quantity tend to rise.

写分析题时把每个箭头改成 because 或 therefore 连接的完整句子；如果中间一步无法解释，说明链条仍有跳跃。

### 数字与情境示范 | Numerical or contextual example

手机价格由 \$500 降至 \$450，购买量 1 000 增至 1 200，是 extension of demand 200。若价格仍为 \$500 而收入增加后购买量变为 1 300，则 demand increases，整条曲线右移。

### English guided explanation

The curve represents the relationship between the product's own price and quantity demanded. Therefore, a change in any other determinant changes the relationship itself and requires a new curve. State both the determinant and why willingness or ability to buy changes.

### Cambridge-style multiple-choice application

Question. The price of petrol rises. What happens to demand for petrol cars, assuming they are complements?

Answer and reasoning. Demand for petrol cars decreases and its demand curve shifts left. The car price need not have changed. 选择题不仅要找到正确项，还要说明其余方向为什么与定义、公式或因果关系不符。

### Cambridge-style structured application

Question. Analyse how higher household income may affect demand for bus travel.

Model development. If bus travel is an inferior service, higher income enables households to switch to cars or taxis. Demand for bus travel falls and its curve shifts left. If buses are high-quality and environmentally preferred, they may instead be normal, so the result depends on consumer behaviour.

### 高频混淆点 | Common misconception

“价格下降使需求增加”在 Cambridge 精确口径中应写 quantity demanded increases 或 extension in demand。

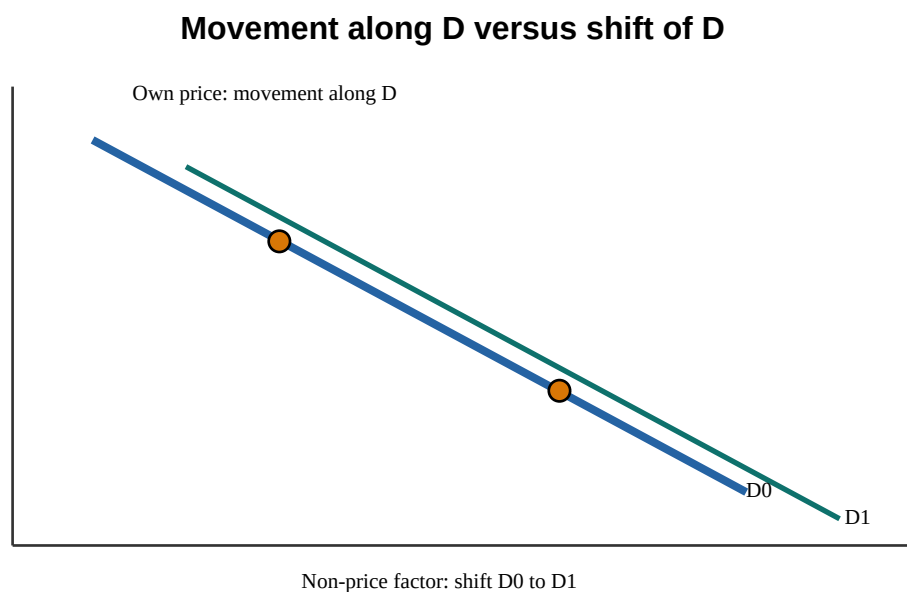
### 条件与评价 | Conditions and evaluation

相关商品关系和 normal/inferior 属性不是永远固定，可能随收入水平、地区、品质与时间改变。

评价不是另起一个无关观点，而是改变时间、幅度、弹性、资源条件或利益相关者后，检查原结论是否仍成立。



## 图表推演 | Diagram analysis



考试图表必须标出坐标轴、曲线名称、新旧均衡和变化方向。文字解释需说明原因、曲线移动以及价格或数量结果。

## IGCSE 大考口径 | Cambridge exam focus

大考高频辨析: own price change = movement along; non-price determinant = shift。必须正确使用 quantity demanded 与 demand。

## 第 9 课 | 2.4 供给与供给决定因素 | Supply and its determinants

所属篇章：资源配置 | The allocation of resources

### 知识点精讲 | Core knowledge

供给 (supply) 是在一定时期、不同价格下厂商愿意并且能够出售的数量。商品自身价格变化造成沿供给曲线移动；生产成本、技术、间接税、补贴、其他产品利润、企业数量、天气、战争和原材料发现等会使整条供给曲线移动。成本下降或技术改善使供给右移；间接税提高单位成本，使供给左移；补贴使供给右移。

English explanation. Supply is the willingness and ability of producers to sell a product at different prices over a period of time. A rise in the product's own price causes an extension in supply. Lower costs, better technology or a subsidy increase supply.

### 深度展开与真题化应用 | In-depth development

Supply 看的是厂商计划出售量

库存很多不等于市场供给一定多；供给指厂商在不同价格下愿意且能够出售的数量。价格上涨通常提高边际收益，使扩产更有吸引力，形成 extension in supply。若生产能力、成本或政策改变，则每个价格下都可能愿意出售更多或更少，曲线才发生 shift。

决定因素的机制不能只背方向

工资、能源和原材料价格下降会降低 unit cost，提高每单位利润，因此 supply increases。技术改善提高 productivity，也会降低 average cost。Indirect tax 提高单位成本，使 S 左移；subsidy 降低有效成本，使 S 右移。好天气提高农作物产量；灾害、战争或资源枯竭限制生产。其他产品更有利可图时，企业可能转移资源，当前产品供给减少。

English development

Supply is the willingness and ability of producers to sell at different prices over a period. A change in the product's own price causes a movement along the supply curve. A change in production costs, technology, taxes, subsidies, weather or the number of firms shifts the curve. The explanation should connect the determinant to cost, profit and planned output.

真题化知识应用

题目：“Explain why a subsidy may increase the supply of electric buses.” 展开为：The subsidy reduces the cost per bus; profit at each market price rises; producers are willing and able to manufacture more; supply shifts right. 只写 “subsidy makes supply rise” 通常缺少分析链。

### 逐题精讲与评价拓展 | Exam workshop

进一步拆解：这一课真正考什么

Supply 必须包含 willingness and ability to sell at different prices over time。Own price 变化导致 extension/contraction in supply；成本、技术、税补贴、天气、企业数或其他产品利润改变 supply curve。

完整因果链 | Cause-and-effect chain

electricity cost falls → unit production cost falls → profit per unit at every price rises → firms are willing and able to sell more → supply shifts right → equilibrium price tends to fall and quantity rises.

写分析题时把每个箭头改成 because 或 therefore 连接的完整句子；如果中间一步无法解释，说明链条仍有跳跃。

## 数字与情境示范 | Numerical or contextual example

价格 \$5 时  $Q_s=60$ 。价格升至 \$7 后同一曲线上  $Q_s=90$ ，是 extension 30。若技术改善后在 \$5 即可供应 100，则 supply increases 40，曲线右移。

## English guided explanation

Supply describes planned sales, not all existing stock. A warehouse may contain goods that the owner is unwilling to sell at the current price. Explain shifts through costs, profitability and productive capacity rather than memorising directions only.

## Cambridge-style multiple-choice application

Question. Which change shifts the supply of bread left: lower flour prices, a subsidy, improved ovens, or a tax on each loaf?

Answer and reasoning. A tax on each loaf, because it raises unit cost and reduces profitability at every market price. 选择题不仅要找到正确项，还要说明其余方向为什么与定义、公式或因果关系不符。

## Cambridge-style structured application

Question. Explain two reasons why the supply of agricultural products may be price inelastic in the short run.

Model development. Crops require time to grow, so farmers cannot quickly raise current harvests when price rises. Available land and storage may also be limited. Hence the percentage change in quantity supplied is smaller than the percentage change in price.

## 高频混淆点 | Common misconception

“供给增加”不能仅由产品自身价格上涨造成；那是 quantity supplied 增加。

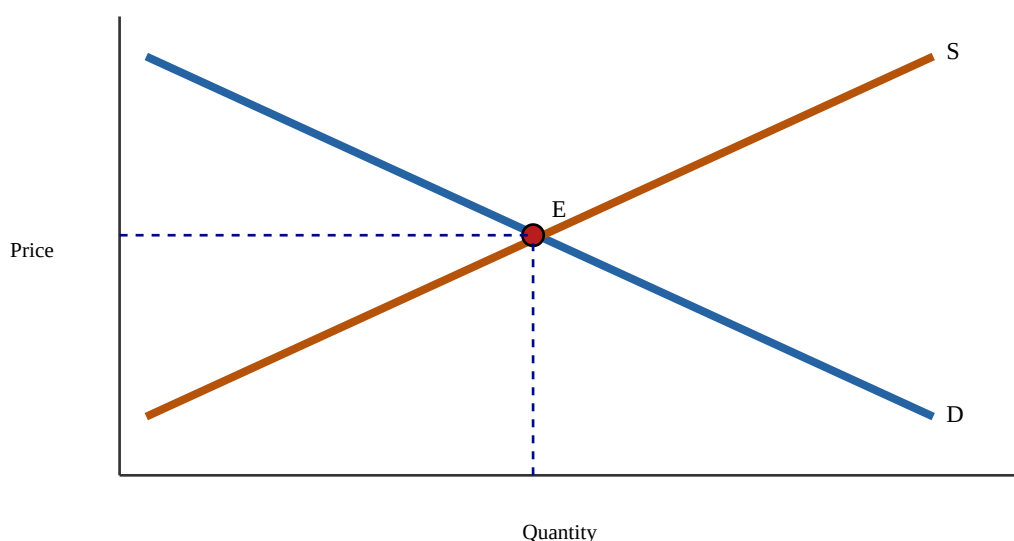
## 条件与评价 | Conditions and evaluation

供给反应随 time period 变强，但易腐性、土地限制、技能和法规可能使长期供给仍缺乏弹性。

评价不是另起一个无关观点，而是改变时间、幅度、弹性、资源条件或利益相关者后，检查原结论是否仍成立。

## 图表推演 | Diagram analysis

### Demand, supply and equilibrium



考试图表必须标出坐标轴、曲线名称、新旧均衡和变化方向。文字解释需说明原因、曲线移动以及价格或数量结果。

## IGCSE 大考口径 | Cambridge exam focus

大考口径：准确定义；紧扣题干应用；解释完整因果链；评价结论说明条件。

# 第 10 课 | 2.5 价格机制、均衡、短缺与过剩 | Price mechanism and equilibrium

所属篇章：资源配置 | The allocation of resources

## 知识点精讲 | Core knowledge

价格机制通过价格变化协调消费者和企业的决定，并回答生产什么、如何生产和为谁生产。均衡出现在 quantity demanded equals quantity supplied。低于均衡价时出现 shortage，买方竞争产生价格上升压力；高于均衡价时出现 surplus，库存增加产生价格下降压力。

English explanation. The price mechanism allocates resources through changes in demand, supply and price. A shortage causes upward pressure on price; a surplus causes downward pressure. The market moves towards equilibrium where planned purchases equal planned sales.

## 深度展开与真题化应用 | In-depth development

### 1. 价格机制到底是什么

价格机制 (price mechanism) 是价格根据 demand 与 supply 的变化而调整，并据此配置稀缺资源的过程。它不是某个人发出命令，而是大量消费者与企业分别作出决定后，通过价格相互协调。完整分析包含三层：外部事件先改变 demand 或 supply；市场随后出现 shortage 或 surplus；价格再发生变化，使 quantity demanded 和 quantity supplied 沿原曲线调整，市场趋向新均衡。

例如某城市突然流行电动自行车，需求曲线右移。在原价格下  $Q_d$  大于  $Q_s$ ，门店很快售罄。短缺向卖方传递“消费者愿意购买更多”的信息，价格可能上涨，企业也更愿意进货或扩产，最终形成较高的 equilibrium price 与较大的 equilibrium quantity。这条链比“需求增加，所以价格上涨”更完整。

### 2. 信号功能 | Signalling function

价格像一个不断更新的信息系统。价格上涨可能表示商品变得相对稀缺，原因可能是需求增加或供给减少；价格下降可能表示需求减弱或供给增加。企业不需要知道每一位消费者为什么改变选择，也能从销量、库存和价格观察市场信号。不过，单看价格不能确定曲线移动原因：同样是 price rise，可能来自 D right，也可能来自 S left，两者对 equilibrium quantity 的影响相反。

English explanation. The signalling function means that changes in price communicate information about relative scarcity. A rising price may signal stronger demand or reduced supply. Producers should also consider whether the change is temporary, the cost of expansion and expected future demand.

### 3. 激励功能 | Incentive function

价格上升通常提高每单位潜在 revenue 和 profit，形成增产激励。现有企业可能增加 workers' hours、使用 spare capacity 或购买 machinery；较长时期内，新企业可能 enter the market。于是 quantity supplied 先沿供给曲线增加，若企业数量或生产能力增长，整条 supply curve 还可能右移。持续低价和亏损则促使企业减产或退出，让 labour、capital 和 land 转向回报更高的用途。

激励并不保证供给立刻大幅增加。鲜花、农作物和住房需要生产时间，PES 较低；制造商若有库存和闲置设备，反应可能更快。高价也可能由间接税造成，并不表示生产者实际收入同比上涨。因此评价要考虑 time period、PES、capacity 和 costs。

English explanation. The incentive function changes behaviour. A higher market price can raise expected profit, encouraging existing firms to expand and new firms to enter. The speed and size of the response depend on price elasticity of supply, spare capacity, stocks and the production period.

### 4. 配给功能 | Rationing function

当商品数量有限，价格上涨会减少 quantity demanded，把产品分配给 willing and able to pay 的消费者。演唱会座位、城市中心住房或干旱时期的水都可能通过价格配给。这里的 able to pay 很重要：市场配给依据购买力，不是依据“谁最需要”。所以价格机制可能有效消除 shortage，却产生 equity 问题；低收入家庭可能买不起必需品。政府可使用 subsidy、direct provision 或 maximum price，但这些政策也可能造成财政成本或新的短缺。

English explanation. The rationing function allocates a scarce product among competing consumers. A higher price reduces quantity demanded, so the available supply goes to those willing and able to pay. This may restore equilibrium, but it does not necessarily produce a fair distribution.

## 5. 价格机制怎样回答三个基本经济问题

生产什么 | What to produce? 消费者通过 demand

表达偏好。需求增加通常使价格与利润上升，企业把资源转入该市场；需求下降则使资源退出。准确说法是 consumer sovereignty works through demand, price and profit signals。

如何生产 | How to produce? 企业比较 labour、capital、land

等要素的相对价格和生产率，选择成本较低且符合质量要求的方法。若工资相对机器价格上升，企业可能采用 capital-intensive production；但机器购置成本、技能、产量规模和质量也会影响决定，所以不能简单写“永远选最便宜要素”。

为谁生产 | For whom to produce? 商品主要流向愿意且能够支付市场价格的人，购买力来自

wages、rent、interest、profit、wealth 或政府转移。市场按 income and purchasing power 分配，而不是按 need 分配，这正是效率与公平之间的重要张力。

English summary. Consumer demand and profit signals influence what is produced. Relative factor prices and productivity influence how it is produced. Income and purchasing power influence for whom it is produced. These outcomes may be efficient, but they are not necessarily equitable.

## 6. 均衡是一组价格与数量

均衡点 E 同时给出 equilibrium price  $P_e$  和 equilibrium quantity  $Q_e$ 。 $P_e$  以下  $Q_d > Q_s$ ，出现 shortage； $P_e$  以上  $Q_s > Q_d$ ，出现 surplus。价格变化使需求量和供给量沿各自曲线移动，直到 planned purchases equal planned sales。均衡不是“公平价格”，也不保证每个人满意；它只表示市场没有内在的价格变化压力。

## 7. 供过于求 | Surplus: $Q_s > Q_d$

假设 equilibrium price 是 \$6、quantity 是 100。若价格为 \$8，消费者计划购买 70，企业计划出售 130，因此 surplus =  $Q_s - Q_d = 60$ 。库存积压使企业降价；价格下降造成 extension in demand 和 contraction in supply，差额逐渐缩小，直到回到均衡。D 与 S 都没有移动，因为引起调整的是商品自身价格。

English chain. Price is above equilibrium → quantity supplied exceeds quantity demanded → unsold stocks accumulate → sellers reduce price → quantity demanded extends and quantity supplied contracts → the surplus disappears at equilibrium.

口诀  $Q_s > Q_d = \text{surplus}$  只负责快速判断；完整答案还要写 unsold stock 和 downward pressure on price。

## 8. 供不应求 | Shortage: $Q_d > Q_s$

若价格为 \$4，消费者计划购买 140，企业只计划出售 60，因此

shortage =  $Q_d - Q_s = 80$ 。消费者竞争、排队或缺货使价格承受上升压力；价格上涨造成 contraction in demand 和 extension in supply，直到  $Q_d = Q_s$ 。若 maximum price 阻止价格上升，shortage 可能持续，并出现 waiting lists、black markets 或 non-price rationing。

English chain. Price is below equilibrium → quantity demanded exceeds quantity supplied → excess demand creates upward pressure on price → quantity demanded contracts and quantity supplied extends → equilibrium is restored.

口诀  $Q_d > Q_s = \text{shortage}$ 。Shortage 不是“供给减少”的同义词，而是某一个价格下  $Q_d$  与  $Q_s$  的差额。

## 9. 用供需数据验证两张图

Price \$8:  $Q_d$  70,  $Q_s$  130 → surplus 60 → downward pressure on price.

Price \$6:  $Q_d$  100,  $Q_s$  100 → equilibrium → no tendency for price to change.

Price \$4:  $Q_d$  140,  $Q_s$  60 → shortage 80 → upward pressure on price.

读表顺序：在同一价格比较  $Q_d$  和  $Q_s$ ；用较大者减较小者计算

gap；最后判断价格方向。不要用不同价格的两行数据相减，也不要写 shortage 写成负数。

## 10. Cambridge 选择题怎样判断

仿真题：At the current price, firms plan to sell 900 units but consumers plan to buy 650 units. What is most likely to happen in a free market? 因为  $900 > 650$ ，所以 surplus=250；unsold stocks 增加，企业最可能 lower price。降价后 quantity demanded increases、quantity supplied decreases。正确选项必须同时符合这条路径。

常见干扰项是 “demand increases”。市场价格下降只带来 extension in demand；题干没有 income、taste 或 substitute price 等非价格因素时，不能写 demand curve shifts right。

## 11. Cambridge 结构题示范

题目：Explain how the price mechanism may remove a shortage of cooking oil.

示范答案：A shortage exists because quantity demanded is greater than quantity supplied at the current price. Consumers compete for the limited cooking oil, creating upward pressure on price. The higher price causes a contraction in quantity demanded because some consumers are no longer willing and able to buy as much. It also causes an extension in quantity supplied because selling the oil becomes more profitable. These movements continue until quantity demanded equals quantity supplied at the equilibrium price.

这段答案依次给出定义、价格方向、消费者反应、生产者反应和均衡结果。若题目要求 analyse，还可以补充：调整速度取决于 PES、库存、市场信息和政府是否设置价格上限。

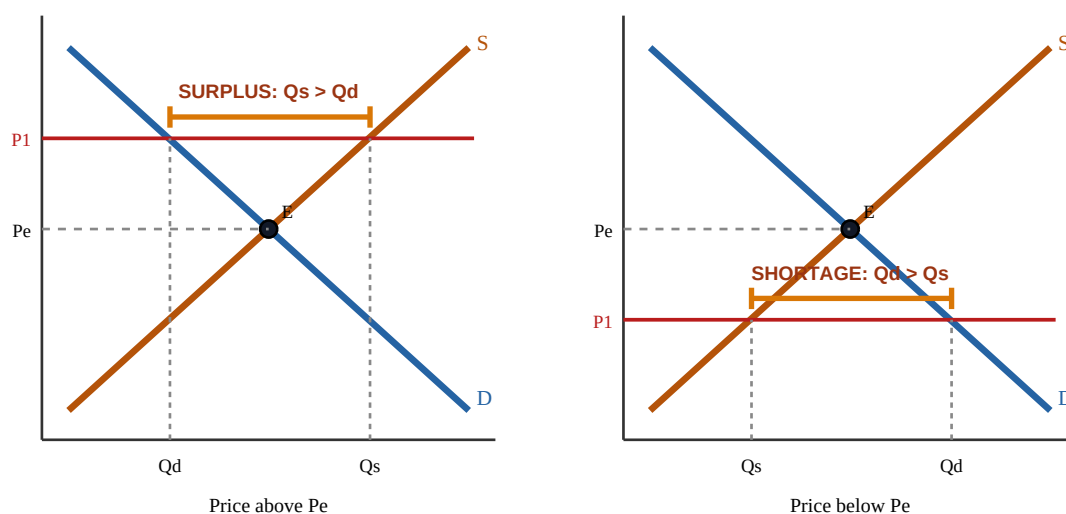
## 12. 图表题的得分细节

两张图都标 Price 纵轴、Quantity 横轴、向下的 D、向上的 S、交点 E 和均衡价  $P_e$ 。Surplus 图在  $P_e$  上方画 P1：左边是  $Q_d$ 、右边是  $Q_s$ ，中间标 surplus。Shortage 图在  $P_e$  下方画 P1：左边是  $Q_s$ 、右边是  $Q_d$ ，中间标 shortage。只画水平线而不标  $Q_d/Q_s$ ，阅卷人无法确认你是否理解差额。



## 图表推演 | Diagram analysis

### Surplus and shortage around equilibrium



考试图表必须标出坐标轴、曲线名称、新旧均衡和变化方向。文字解释需说明原因、曲线移动以及价格或数量结果。

## IGCSE 大考口径 | Cambridge exam focus

大考高频：解释 signalling、incentive、rationing；用  $Q_s > Q_d$  判断 surplus、 $Q_d > Q_s$  判断 shortage；价格调整只造成沿曲线移动。图表必须标 D、S、 $P_e$ 、 $P_1$ 、 $Q_d$ 、 $Q_s$  和缺口。

## 第 11 课 | 2.6 需求和供给同时移动 | Simultaneous changes

所属大章: 资源配置 | The allocation of resources

### 知识点精讲 | Core knowledge

当需求和供给同时右移时，均衡数量一定增加，但均衡价格方向取决于两条曲线移动的相对幅度。如果需求增加幅度大于供给增加，价格上升；若供给增加幅度更大，价格下降。若一条右移、一条左移，则可能只能确定价格或数量中的一个。

English explanation. When both demand and supply increase, equilibrium quantity rises, but the effect on equilibrium price is uncertain. It depends on the relative size of the two shifts. State what is certain and what is indeterminate.

### 深度展开与真题化应用 | In-depth development

先单独判断，再合并

同时移动题不能凭直觉直接下结论。先只看 demand shift 对 P、Q 的作用，再只看 supply shift，最后比较方向。D ↑ 使 P ↑ Q ↑；S ↑ 使 P ↓ Q ↑，合并后 Q 一定 ↑，P 不确定。D ↓ 与 S ↓ 使 Q 一定 ↓，P 不确定。D ↑ 与 S ↓ 使 P 一定 ↑，Q 不确定；D ↓ 与 S ↑ 使 P 一定 ↓，Q 不确定。

“不确定”也需要经济解释

不确定不是不会做，而是缺少 relative magnitude。若 demand increase 比 supply increase 大，价格上升；若 supply increase

更大，价格下降；幅度相同则价格可能不变。图中可以画两种可能的新交点，或在文字中清楚说明条件。

English development

Analyse simultaneous shifts in two stages. State the separate effect of each shift, identify the variable on which both effects agree, and explain why the other variable is indeterminate. Never claim that both price and quantity are uncertain without checking the direction of each individual effect.

真题化知识应用

例如健康报告提高 tomatoes

的需求，同时丰收提高供给。均衡数量一定增加；价格取决于需求与供给右移的相对幅度。结构题得分点包括 D right、S right、Q increases、P uncertain，以及 because the relative size of the shifts is unknown。

### 逐题精讲与评价拓展 | Exam workshop

进一步拆解：这一课真正考什么

同时移动题先建立四格方向表，再看两种作用是否同向。D ↑ S ↑ 时 Q 同向增加、P 方向冲突；D ↑ S ↓ 时 P 同向上升、Q 冲突。能够判断“不确定”是分析结果，不是信息遗漏。

完整因果链 | Cause-and-effect chain

healthy diet campaign → tomato demand right → P ↑ Q ↑；good harvest → tomato supply right → P ↓ Q ↑；合并后 Q ↑，P depends on relative shifts。写分析题时把每个箭头改成 because 或 therefore 连接的完整句子；如果中间一步无法解释，说明链条仍有跳跃。

数字与情境示范 | Numerical or contextual example

原均衡 P=10, Q=100。若需求变化单独使 P=13, Q=120，供给变化单独使 P=8, Q=125，合并后 Q 必高于100，但没有曲线幅度就不能把 13 与 8 平均为 10.5。

## English guided explanation

Do not add or average separate equilibrium prices. The final equilibrium is the intersection of the two new curves. State the certain variable first and then explain why the other is indeterminate.

## Cambridge-style multiple-choice application

Question. Demand falls while supply rises. Which outcome is certain?

Answer and reasoning. Equilibrium price falls. Both shifts put downward pressure on price, while their effects on quantity oppose each other.

选择题不仅要找到正确项，还要说明其余方向为什么与定义、公式或因果关系不符。

## Cambridge-style structured application

Question. Analyse the effect on the market for electric cars of a successful advertising campaign and a fall in battery costs.

Model development. Advertising shifts demand right. Lower battery costs shift supply right. Both changes increase equilibrium quantity. Demand growth raises price while supply growth lowers it, so the final price depends on the relative size of the shifts.

## 高频混淆点 | Common misconception

不要把两条曲线右移理解为 price 和 quantity 都必然上升。必须分别追踪每条移动。

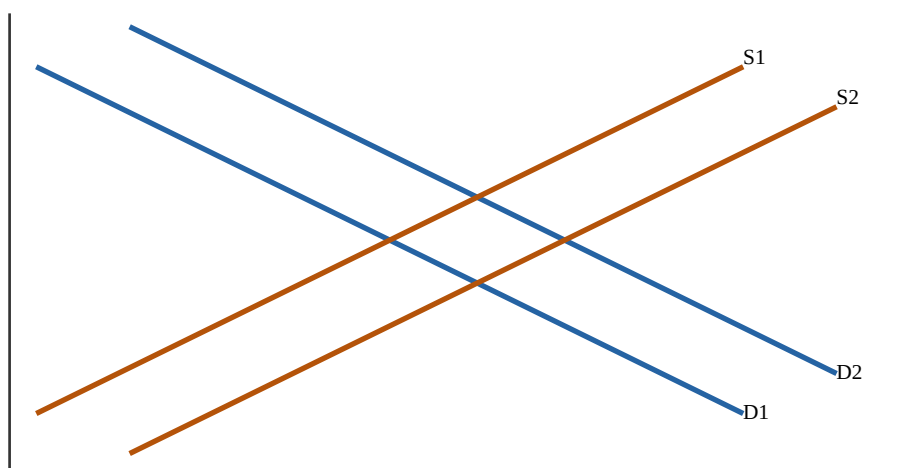
## 条件与评价 | Conditions and evaluation

若题干提供幅度、弹性或新交点，价格可能确定；没有信息时应明确写 indeterminate，而非猜测。

评价不是另起一个无关观点，而是改变时间、幅度、弹性、资源条件或利益相关者后，检查原结论是否仍成立。

## 图表推演 | Diagram analysis

### Simultaneous changes in demand and supply



Quantity rises; price depends on relative shifts

考试图表必须标出坐标轴、曲线名称、新旧均衡和变化方向。文字解释需说明原因、曲线移动以及价格或数量结果。

## IGCSE 大考口径 | Cambridge exam focus

大考图形题：先分别移动 D、S，再比较新旧均衡；明确写出哪些变量可以确定，哪些取决于曲线移动幅度。

## 第 12 课 | 2.7 PED、PES 与销售收入 | Elasticity and revenue

所属大章: 资源配置 | The allocation of resources

### 知识点精讲 | Core knowledge

$PED = \text{percentage change in quantity demanded} \div \text{percentage change in price}$ 。绝对值大于 1 为 elastic, 小于 1 为 inelastic, 等于 1 为 unit elastic。对缺乏弹性的需求, 提高价格会使 total revenue 上升; 对富有弹性的需求, 提高价格会使 total revenue 下降。PES 取决于闲置产能、库存、生产周期和要素流动性。

English explanation. Price elasticity measures responsiveness, not the direction of a curve shift. Always show percentage changes and interpret the value in context. A firm facing elastic demand may raise revenue by lowering price.

### 深度展开与真题化应用 | In-depth development

#### 弹性衡量 “反应有多大”

PED 使用百分比而非绝对变化, 使不同货币和数量单位可以比较。通常取绝对值分类:  $|PED| > 1$  elastic,  $|PED| < 1$  inelastic。必需品、替代品少、支出占比低以及短期通常使需求较缺乏弹性; 奢侈品、替代品多和较长时间使需求较富弹性。PES 则取决于 spare capacity、stocks、production period 和 factor mobility。

#### PED 与 total revenue

$TR = P \times Q$ 。若需求缺乏弹性, 价格提高的百分比大于数量减少的百分比, TR 上升; 若需求富有弹性, 数量减少比例更大, TR 下降。计算时例如 P 由 10 增至 12 (20%), Q 由 100 降至 70 (30%),  $PED = 30/20 = 1.5$ , 属 elastic, TR 从 1000 降至 840。

#### English development

Price elasticity of demand measures the responsiveness of quantity demanded to a change in price. It does not explain why demand shifts. Price elasticity of supply measures how easily producers can change output when price changes. Interpretation must include both the numerical category and its meaning in context.

#### 真题化知识应用

选择题可能不要求算公式, 而用 revenue 判断: price rises and total revenue falls implies elastic demand。问答题 “Explain two causes of inelastic demand for rice” 应发展 necessity、few close substitutes、small budget share 或 short time period, 每点都说明消费者为何难以减少购买量。

### 逐题精讲与评价拓展 | Exam workshop

#### 进一步拆解: 这一课真正考什么

弹性比较百分比反应。PED 通常为负, 但分类常用绝对值; PES 通常为正。计算后必须用 elastic/inelastic/unit elastic 分类, 并解释 “价格变 1%, 数量大约变多少”。

#### 完整因果链 | Cause-and-effect chain

many close substitutes  $\rightarrow$  consumers can switch easily after price rise  $\rightarrow$  large percentage fall in  $Q_d \rightarrow$  PED more elastic。spare capacity  $\rightarrow$  firms expand quickly after price rise  $\rightarrow$  larger percentage rise in  $Q_s \rightarrow$  PES more elastic。写分析题时把每个箭头改成 because 或 therefore 连接的完整句子; 如果中间一步无法解释, 说明链条仍有跳跃。

#### 数字与情境示范 | Numerical or contextual example

价格由 \$20 降至 \$18，下降10%；Qd由200增至250，增加25%； $|PED|=25/10=2.5$ ，elastic。TR 由4000增至4500，符合降价提高富弹性需求下收入。

### English guided explanation

Elasticity measures responsiveness, not slope alone. The same straight demand curve has different elasticity at different points. Determinants include substitutes, necessity, budget share and time; for supply, capacity, stock and production time are central.

### Cambridge-style multiple-choice application

Question. Price rises by 8% and quantity demanded falls by 4%. What is PED and what happens to total revenue?

Answer and reasoning. PED is 0.5 in absolute value, so demand is inelastic and total revenue is likely to rise. 选择题不仅要找到正确项，还要说明其余方向为什么与定义、公式或因果关系不符。

### Cambridge-style structured application

Question. Analyse why demand for a life-saving medicine may be price inelastic and how this affects a tax.

Model development. The medicine is a necessity and may have few substitutes, so patients reduce purchases by a smaller percentage than the price rise. A tax can therefore raise substantial revenue, but consumption and the related external cost may fall only slightly. Low-income patients may bear a large burden.

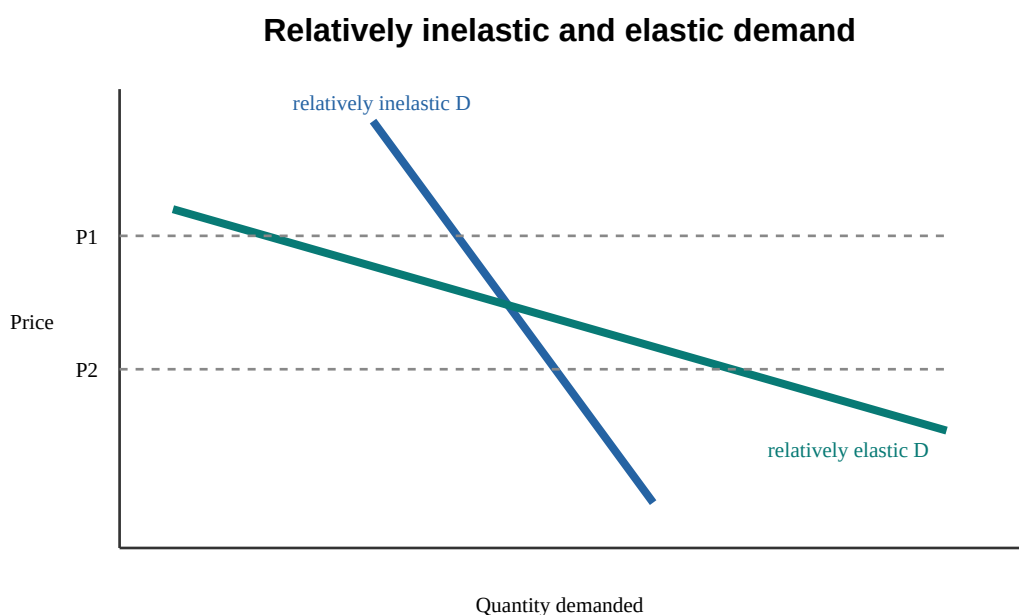
### 高频混淆点 | Common misconception

弹性没有货币单位。不要把“2”解释为数量增加2个；应写 quantity changes by about 2% for a 1% price change。

### 条件与评价 | Conditions and evaluation

企业收入结论要求其他条件不变。竞争者反应、品牌差异和成本变化可能让实际 revenue 与简单 PED 预测不同。评价不是另起一个无关观点，而是改变时间、幅度、弹性、资源条件或利益相关者后，检查原结论是否仍成立。

## 图表推演 | Diagram analysis



考试图表必须标出坐标轴、曲线名称、新旧均衡和变化方向。文字解释需说明原因、曲线移动以及价格或数量结果。

## IGCSE 大考口径 | Cambridge exam focus

大考口径：准确定义；紧扣题干应用；解释完整因果链；评价结论说明条件。

## 第 13 课 | 2.8 市场失灵与政府干预 | Market failure and intervention

所属大章: 资源配置 | The allocation of resources

### 知识点精讲 | Core knowledge

市场失灵指自由市场没有把资源配置到社会最优位置, 原因包括 external costs and benefits、public goods、information failure 和 market power。政策包括 indirect taxes、subsidies、regulation、information provision、price controls 和 direct provision。

English explanation. Market failure occurs when the price mechanism does not allocate resources efficiently. A tax may reduce consumption of a demerit good, but effectiveness depends on PED, enforcement costs and unintended effects.

### 深度展开与真题化应用 | In-depth development

#### 从私人决定到社会结果

市场失灵发生在 free market allocation 不能最大化社会福利。负外部性中, 生产者或消费者只考虑 private cost, 却把污染、噪声或健康损害转嫁给 third parties, 因此  $\text{social cost} > \text{private cost}$ , 市场数量可能过多。正外部性中  $\text{social benefit} > \text{private benefit}$ , 教育或疫苗可能被少消费。

#### 不同失灵需要不同政策

Public goods 具有 non-excludability 和 non-rivalry, free-rider problem 使私人企业难收费, 政府可能直接提供。Information failure 使消费者低估有益品或高估有害品。Indirect tax、subsidy、regulation、information campaign、maximum/minimum price 和 direct provision 各有成本、时滞与副作用。

#### English development

Market failure is a misallocation of resources. An indirect tax can internalise an external cost by raising private cost, shifting supply left and reducing output. Its success depends on the size of the tax, PED, enforcement and whether producers or consumers can avoid the policy. Government intervention may itself be imperfect.

#### 真题化知识应用

讨论 carbon tax 时, 支持面应写 price rises  $\rightarrow$  quantity demanded falls  $\rightarrow$  emissions/external cost falls, 并可加 tax revenue; 反面写 inelastic demand、job losses、administrative cost、regressive impact 或 international competitiveness。最后依据污染严重程度、替代能源可得性和收入用途判断。

### 逐题精讲与评价拓展 | Exam workshop

#### 进一步拆解: 这一课真正考什么

市场失灵要定位 missing cost/benefit、missing market、missing information 或 excessive market power。用 private 与 social 视角解释为何 free-market quantity 偏离 socially efficient quantity, 再选择针对性政策。

#### 完整因果链 | Cause-and-effect chain

factory pollution  $\rightarrow$  external cost imposed on residents  $\rightarrow$  private cost below social cost  $\rightarrow$  market price too low/output too high  $\rightarrow$  indirect tax raises private cost  $\rightarrow$  supply left  $\rightarrow$  output and pollution may fall。  
写分析题时把每个箭头改成 because 或 therefore 连接的完整句子; 如果中间一步无法解释, 说明链条仍有跳跃。



## 数字与情境示范 | Numerical or contextual example

每单位产品 private cost=\$6, 污染 external cost=\$3, 则 social cost=\$9。若市场只按 \$6 作决定, 会忽略 \$3 的第三方损害; 理论上每单位 \$3 的税可使决策者面对更完整成本。

## English guided explanation

An externality affects a third party not directly involved in the transaction. Public goods are non-excludable and non-rival, creating a free-rider problem. Information failure occurs when choices are based on inaccurate or incomplete information.

## Cambridge-style multiple-choice application

Question. Which is the clearest example of an external cost: a firm pays wages, a smoker harms nearby non-smokers, a consumer buys food, or a bank charges interest?

Answer and reasoning. Harm to nearby non-smokers, because the cost falls on third parties outside the transaction. 选择题不仅要找到正确项, 还要说明其余方向为什么与定义、公式或因果关系不符。

## Cambridge-style structured application

Question. Discuss whether a tax is the best way to reduce consumption of sugary drinks.

Model development. A tax raises firms' costs and price, reducing quantity demanded and health external costs while raising revenue. However, demand may be inelastic, the tax may be regressive and substitutes may also be unhealthy. Information campaigns or regulation may complement the tax. Effectiveness depends on tax size and alternatives.

## 高频混淆点 | Common misconception

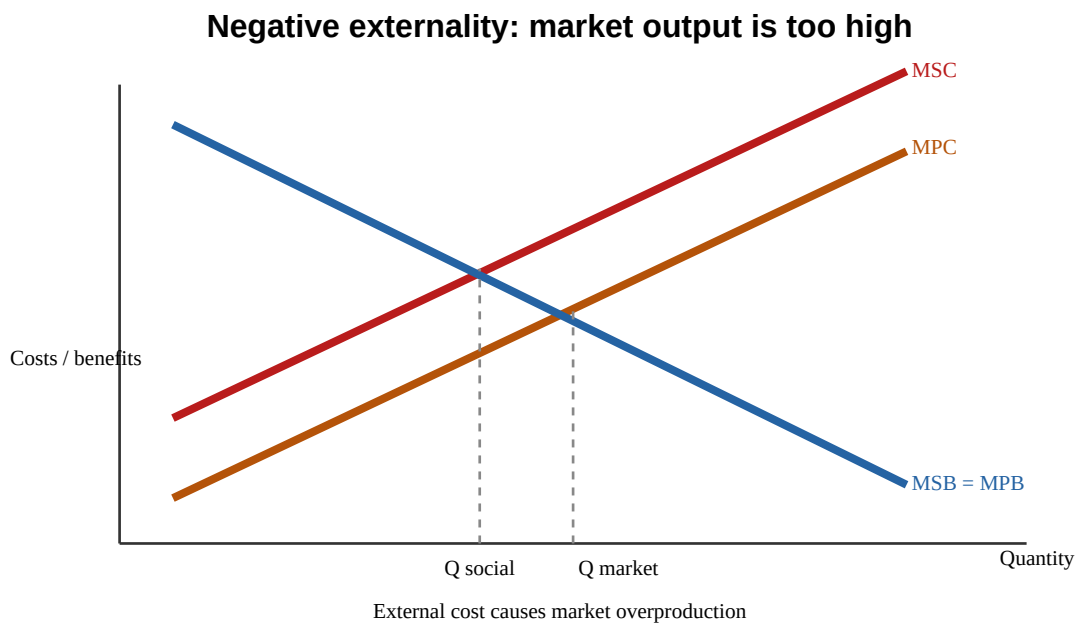
Public good 不等于“政府提供的好东西”。必须同时具有 non-excludability 与 non-rivalry。

## 条件与评价 | Conditions and evaluation

政府干预也有 information、administrative cost、unintended consequence 和 political pressure, 评价时比较 market failure 与 government failure。

评价不是另起一个无关观点, 而是改变时间、幅度、弹性、资源条件或利益相关者后, 检查原结论是否仍成立。

## 图表推演 | Diagram analysis



考试图表必须标出坐标轴、曲线名称、新旧均衡和变化方向。文字解释需说明原因、曲线移动以及价格或数量结果。

## IGCSE 大考口径 | Cambridge exam focus

大考口径：准确定义；紧扣题干应用；解释完整因果链；评价结论说明条件。

## 第 14 课 | 2.1 货币与银行 ( Money and banking )

所属大章: 微观经济决策者 | Microeconomic decision makers

### 知识点精讲 | Core knowledge

货币的四项功能是 medium of exchange ( 交换媒介 )、measure of value ( 价值尺度 )、store of value ( 价值储藏 ) 与 standard of deferred payment ( 延期支付标准 )。商业银行 ( commercial bank ) 接受存款、提供支付服务并放贷; 中央银行 ( central bank ) 通常发行货币、监管银行并影响利率。通胀削弱货币的购买力, 因此“钱可以储藏价值”不等于“价值永远不变”。

### 深度展开与真题化应用 | In-depth development

#### 货币功能与良好货币特征

货币作为 medium of exchange 消除 barter 的 double coincidence of wants; 作为 measure of value 让商品可比较; 作为 store of value 把购买力转移到未来; 作为 standard of deferred payment 支持借贷。良好货币应 durable、portable、divisible、recognisable、scarce and stable in value。高通胀会削弱价值储藏功能。

#### 商业银行与中央银行

商业银行接受 deposits、提供 loans、支付和外汇服务, 并通过利差盈利。中央银行通常发行货币、监管银行、作为政府银行和最后贷款人, 并实施 monetary policy。利率上升提高借款成本, 可能减少消费和投资; 同时提高储蓄回报。结果取决于信心、债务类型和利率变化幅度。

#### English development

Commercial banks connect savers with borrowers and provide liquidity and payment services. A central bank is responsible for monetary stability and supervision. Do not confuse money with income or wealth: money is an asset used to make transactions, while income is a flow received over time.

#### 真题化知识应用

题目若问 “Explain two functions of money”, 每个功能要说明如何发挥作用。若问利率上升对家庭, 既可写 mortgage repayments rise → disposable income falls → consumption falls, 也可写 saving becomes more rewarding → saving rises。

### 逐题精讲与评价拓展 | Exam workshop

#### 进一步拆解: 这一课真正考什么

先区分 money、income、wealth 和 credit。货币是被普遍接受的交易资产; 收入是一段时间的流量; 财富是资产存量; 信用是借款能力。再区分 commercial bank 与 central bank。

#### 完整因果链 | Cause-and-effect chain

central bank raises interest rate → borrowing becomes dearer/saving more attractive → household consumption and firm investment may fall → AD falls → inflationary pressure may ease。

写分析题时把每个箭头改成 because 或 therefore 连接的完整句子; 如果中间一步无法解释, 说明链条仍有跳跃。

#### 数字与情境示范 | Numerical or contextual example

存款 \$1 000、年利率3%, 一年利息约 \$30。若通胀5%, 实际购买力仍可能下降约2%, 说明 nominal return 与 real return 不同。

## English guided explanation

Money performs four functions: medium of exchange, measure of value, store of value and standard of deferred payment. Commercial banks accept deposits and make loans. The central bank conducts monetary policy, supervises the banking system and may act as lender of last resort.

## Cambridge-style multiple-choice application

Question. Which function of money is weakened most directly by rapid inflation?

Answer and reasoning. Store of value, because money loses purchasing power quickly over time.

选择题不仅要找到正确项，还要说明其余方向为什么与定义、公式或因果关系不符。

## Cambridge-style structured application

Question. Analyse how a rise in interest rates may affect households and firms.

Model development. Loan and mortgage repayments may rise, reducing disposable income and consumption. Saving becomes more attractive. Firms face a higher cost of borrowing, so fewer investment projects are profitable. Demand and employment may fall, although savers receive more interest.

## 高频混淆点 | Common misconception

中央银行通常不为普通家庭办理日常存款；商业银行也不负责制定全国政策利率。

## 条件与评价 | Conditions and evaluation

利率传导取决于固定或浮动贷款比例、信心、负债水平和时滞，不能断言消费投资立即下降。

评价不是另起一个无关观点，而是改变时间、幅度、弹性、资源条件或利益相关者后，检查原结论是否仍成立。

## 英文精讲 | English explanation

Money is a medium of exchange, a measure of value, a store of value and a standard of deferred payment.

## IGCSE 大考口径 | Cambridge exam focus

大考口径：准确定义；紧扣题干应用；解释完整因果链；评价结论说明条件。

# 第 15 课 | 2.2 家庭、劳动者与工会 (Households, workers and trade unions)

所属大章: 微观经济决策者 | Microeconomic decision makers

## 知识点精讲 | Core knowledge

家庭提供 labour、land、capital，获得收入后在消费与储蓄之间选择。劳动需求是 derived demand（派生需求）：企业需要工人，是因为消费者需要工人生产的商品。工资、技能、资格、工作条件、歧视和工会谈判力量都会影响工资差异。工会可用集体谈判提高工资和安全，也可能在工资高于均衡时增加失业；要写“可能”，不能绝对化。

## 深度展开与真题化应用 | In-depth development

### 家庭如何分配收入

家庭收入可用于 consumption、saving 和 direct taxes。收入越高，通常能储蓄的绝对额越大，但比例还受利率、信心、年龄、财富和未来预期影响。消费者选择也受 price、income、advertising、quality 和 tastes 影响。借贷让当前消费增加，但未来必须偿还本息。

### 工资差异与劳动市场

工资由劳动需求与供给共同影响。需要高技能、培训期长、风险大或工作条件差的职业可能工资较高；工会、最低工资、歧视和公共部门政策也会改变工资。Trade union 通过 collective bargaining 改善工资和条件，但工资高于生产率增长可能增加成本、价格或失业风险。

### English development

Demand for labour is derived from demand for the product workers produce. A rise in labour productivity may increase firms' willingness to pay higher wages because each worker adds more output. Non-wage factors such as hours, job security, promotion and location also affect occupational choice.

### 真题化知识应用

讨论最低工资时，正面可写 higher income、motivation/productivity、lower labour turnover；反面写 higher labour cost → firms reduce employment or raise prices。结论取决于最低工资相对均衡工资的高度、劳动需求弹性和企业能否把成本转嫁给消费者。

## 逐题精讲与评价拓展 | Exam workshop

### 进一步拆解：这一课真正考什么

家庭同时是 consumers、savers、borrowers 和 suppliers of labour。劳动市场选择受 wage 与 non-wage factors 影响；工资差异来自技能、培训、风险、责任、劳动供需、工会和歧视。

### 完整因果链 | Cause-and-effect chain

product demand rises → derived demand for labour rises → labour shortage strengthens worker bargaining power → wages may rise → firms' costs may increase, unless productivity also rises.

写分析题时把每个箭头改成 because 或 therefore 连接的完整句子；如果中间一步无法解释，说明链条仍有跳跃。

### 数字与情境示范 | Numerical or contextual example

月收入 \$2 000，direct tax \$300，消费 \$1 400，则 saving=\$300。利率上升不保证储蓄必然增加：income effect 与 substitution effect 可能方向不同。

### English guided explanation

Labour demand is derived because firms employ workers to produce goods and services. Workers compare wages with hours, conditions, security, promotion and location. Trade unions use collective bargaining and may also provide legal or training services.

### Cambridge-style multiple-choice application

Question. Which change is most likely to increase demand for chefs: lower demand for restaurant meals, higher chef productivity, more people entering chef training, or lower food prices with no sales change?

Answer and reasoning. Higher chef productivity, because each chef contributes more output and revenue to the firm. 选择题不仅要找到正确项，还要说明其余方向为什么与定义、公式或因果关系不符。

### Cambridge-style structured application

Question. Discuss whether trade-union action always benefits workers.

Model development. Collective bargaining may raise wages, improve safety and reduce exploitation. Higher pay can motivate workers and reduce turnover. However, large wage increases can raise costs, encourage automation or reduce employment. Non-members and consumers may lose. The result depends on productivity and labour-demand elasticity.

### 高频混淆点 | Common misconception

Labour supply increase 是更多工人愿意提供劳动; employment increase 是实际受雇人数增加，两者不是同义词。

### 条件与评价 | Conditions and evaluation

工资高低还应与 real wage、工作时长、生活成本、福利和就业稳定性一起判断。

评价不是另起一个无关观点，而是改变时间、幅度、弹性、资源条件或利益相关者后，检查原结论是否仍成立。

## 英文精讲 | English explanation

Households allocate income between consumption, saving and taxation. Workers consider wages and non-wage factors.

## IGCSE 大考口径 | Cambridge exam focus

大考口径：准确定义；紧扣题干应用；解释完整因果链；评价结论说明条件。

## 第 16 课 | 2.3 企业的生产、成本与收益

所属大章：微观经济决策者 | Microeconomic decision makers

### 知识点精讲 | Core knowledge

Fixed costs (租金、厂房) 不随短期产量变化; variable costs (原料、按件工资) 会变化。Total cost = fixed cost + variable cost。Total revenue = price  $\times$  quantity; profit = total revenue - total cost。企业不只追求利润, 也可能追求 sales revenue、market share、survival 或环境声誉; 题目给了目标才应据此分析。

### 深度展开与真题化应用 | In-depth development

#### 生产率、分工与生产阶段

生产率是单位投入产出, 例如 output per worker, 不等于总产量。Division of labour 把生产拆成任务, 使工人 specialised, 减少转换时间并便于使用机器; 但重复工作可能降低 motivation, 停工也会影响整条生产线。Primary、secondary、tertiary sectors 分别涉及资源提取、制造建筑和服务。

#### 成本收益必须按公式连接

FC 不随 output 改变, VC 随 output 改变;  $TC=FC+VC$ ,  $AC=TC/Q$ 。TR= $P \times Q$ , profit=TR-TC。若产量由 100 增至 200, TC 从 1000 增至 1600, AC 从 10 降至 8, 说明平均成本下降, 即使总成本上升。利润变化必须同时看 price、quantity 和 cost。

#### English development

Fixed cost exists even at zero output, while variable cost changes with output. Average cost is not the extra cost of one more unit. Productivity improvements can reduce unit cost, increase competitiveness and raise profit, but new machinery may involve high initial expenditure.

#### 真题化知识应用

成本表题先求  $VC=TC-FC$ , 再求  $AC=TC/Q$ 。解释自动化时写 machinery raises output per worker  $\rightarrow$  unit labour cost may fall  $\rightarrow$  firm can lower price or gain profit; 评价还可写 capital cost、training、job losses 和是否有足够 demand。

### 逐题精讲与评价拓展 | Exam workshop

#### 进一步拆解: 这一课真正考什么

本课连接 production、specialisation、productivity、cost、revenue 与 profit。学生需能由现实变化一路推到单位成本和竞争力, 而不只是背公式。

#### 完整因果链 | Cause-and-effect chain

division of labour  $\rightarrow$  repeated practice and less switching time  $\rightarrow$  output per worker rises  $\rightarrow$  unit cost may fall  $\rightarrow$  price may fall or profit rise  $\rightarrow$  sales/investment may increase。写分析题时把每个箭头改成 because 或 therefore 连接的完整句子; 如果中间一步无法解释, 说明链条仍有跳跃。

#### 数字与情境示范 | Numerical or contextual example

$Q=100$ ,  $FC=\$400$ ,  $VC=\$600$ , 所以  $TC=\$1\,000$ ,  $AC=\$10$ 。若 price= $\$14$ , TR= $\$1\,400$ , profit= $\$400$ 。Q 增加后总成本可上升而平均成本下降。

#### English guided explanation



Production is the transformation of inputs into output. Division of labour can raise productivity through specialisation, but repetitive work may reduce motivation. Profit is a residual: total revenue minus total cost, not the same as sales revenue.

### Cambridge-style multiple-choice application

Question. A firm's output doubles while total cost rises by 50%. What happens to average cost?

Answer and reasoning. Average cost falls, because cost rises less proportionately than output.

选择题不仅要找到正确项，还要说明其余方向为什么与定义、公式或因果关系不符。

### Cambridge-style structured application

Question. Analyse two ways improved labour productivity may affect a firm.

Model development. More output per worker can reduce unit labour cost, allowing a lower price and greater competitiveness. Alternatively, the firm may keep price unchanged and earn more profit, financing investment. The effect depends on wages, product quality and demand.

### 高频混淆点 | Common misconception

FC 不随 output 改变，但 per-unit fixed cost 会变；VC 随 output 变，不代表每单位 VC 必须恒定。

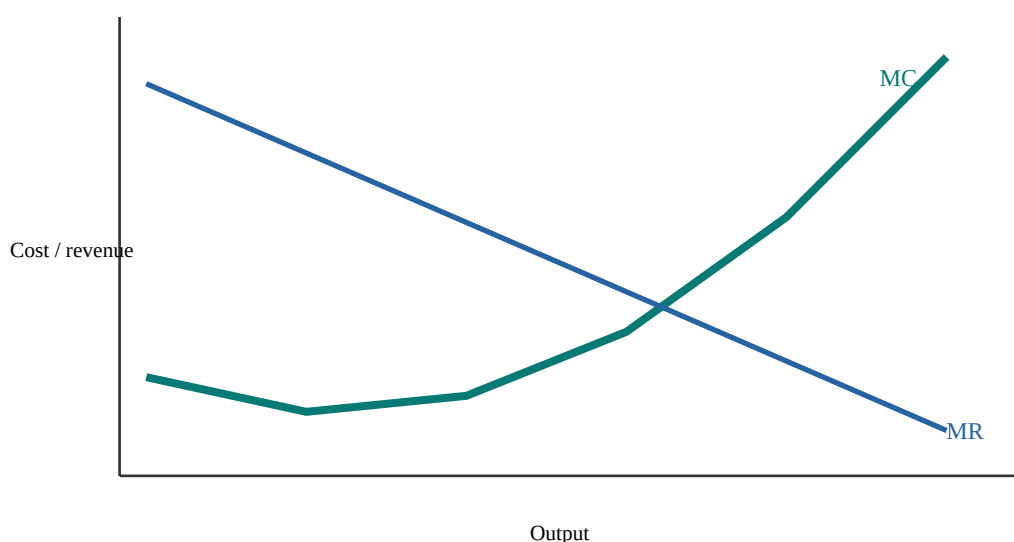
### 条件与评价 | Conditions and evaluation

自动化带来的 productivity 增长要扣除购置、维护和培训成本，并考虑产品需求是否足以吸收新增产量。

评价不是另起一个无关观点，而是改变时间、幅度、弹性、资源条件或利益相关者后，检查原结论是否仍成立。

## 图表推演 | Diagram analysis

### Cost and revenue relationships



考试图表必须标出坐标轴、曲线名称、新旧均衡和变化方向。文字解释需说明原因、曲线移动以及价格或数量结果。

### 英文精讲 | English explanation

Total cost equals fixed cost plus variable cost; average cost is total cost divided by output. Profit equals total revenue minus total cost.

### IGCSE 大考口径 | Cambridge exam focus

大考口径：准确定义；紧扣题干应用；解释完整因果链；评价结论说明条件。

## 第 17 课 | 2.4 规模经济与市场结构

所属大章: 微观经济决策者 | Microeconomic decision makers

### 知识点精讲 | Core knowledge

长期扩张时, 平均成本下降叫 economies of

scale: 可以来自批量采购、专业化、管理、技术和融资。规模太大可能产生 communication problems 和 coordination problems, 即 diseconomies of scale。市场结构从竞争强到弱大致为 perfect

competition、monopolistic competition、oligopoly、monopoly; 答题不要只背定义, 应把进入壁垒、产品差异、定价能力和消费者选择联系起来。

### 深度展开与真题化应用 | In-depth development

#### 企业变大为何可能先降成本

Internal economies of scale 包括 purchasing、technical、financial、managerial、marketing 和 risk-bearing economies。大型企业批量采购可议价, 专用机器产量大时才划算, 借款风险较低, 专业经理提高决策质量, 广告成本可分摊到更多产品。它们影响 long-run average cost, 不等同于短期固定成本分摊。

#### 规模不经济与竞争程度

企业过大可能出现 communication failure、协调困难、管理层级增加和员工疏离, 使 average cost 上升。竞争市场中 firms 多、进入较易, 企业面临降价和创新压力; monopoly 有规模优势和研发资金, 也可能提高价格、限制选择或降低效率。市场份额高不自动证明滥用市场力量。

#### English development

Economies of scale are reductions in average cost resulting from an increase in the scale of production. A monopoly is a dominant or sole supplier, but its effects depend on contestability, regulation and whether scale economies are passed on to consumers.

#### 真题化知识应用

讨论 “more competition benefits consumers” 时, 写 lower prices、better quality、more choice、innovation; 另一面可写 firms lose scale economies、duplicate fixed costs、profit for R&D falls。结论要结合市场是否自然垄断、消费者能否获得信息和监管质量。

### 逐题精讲与评价拓展 | Exam workshop

#### 进一步拆解: 这一课真正考什么

先区分 growth of firm、economies of scale 与 monopoly。规模扩大可能使 long-run average cost 下降, 但超过适合规模又可能出现 diseconomies。市场集中度高影响 competition, 却不自动确定价格、质量或创新结果。

#### 完整因果链 | Cause-and-effect chain

bulk purchasing → stronger bargaining power → lower input price per unit → average cost falls → firm can lower price → market share may rise → further scale economies may develop。写分析题时把每个箭头改成 because 或 therefore 连接的完整句子; 如果中间一步无法解释, 说明链条仍有跳跃。

#### 数字与情境示范 | Numerical or contextual example

小企业年广告费 \$20 000、销量10 000, 广告成本每件\$2; 大企业广告费\$100 000、销量100 000, 每件\$1。这是 marketing economy, 但总广告费更高。

## English guided explanation

Internal economies arise from the firm's own expansion; external economies arise when the industry or region expands. Diseconomies often result from communication, coordination and motivation problems in very large organisations.

## Cambridge-style multiple-choice application

Question. Which is an internal economy of scale: lower national interest rates, a skilled regional labour pool, bulk-buying discounts, or a new public road?

Answer and reasoning. Bulk-buying discounts, because they result directly from the firm's larger purchasing scale. 选择题不仅要找到正确项，还要说明其余方向为什么与定义、公式或因果关系不符。

## Cambridge-style structured application

Question. Discuss whether consumers benefit from a monopoly.

Model development. A monopoly may exploit scale economies, fund research and provide a standardised network. It may pass lower costs to consumers. Conversely, weak competition may permit higher prices, less choice and lower quality. Outcomes depend on regulation, contestability and the source of monopoly power.

## 高频混淆点 | Common misconception

规模经济是 average cost 下降，不是 total cost 下降；market share 大也不等于只有一家企业。

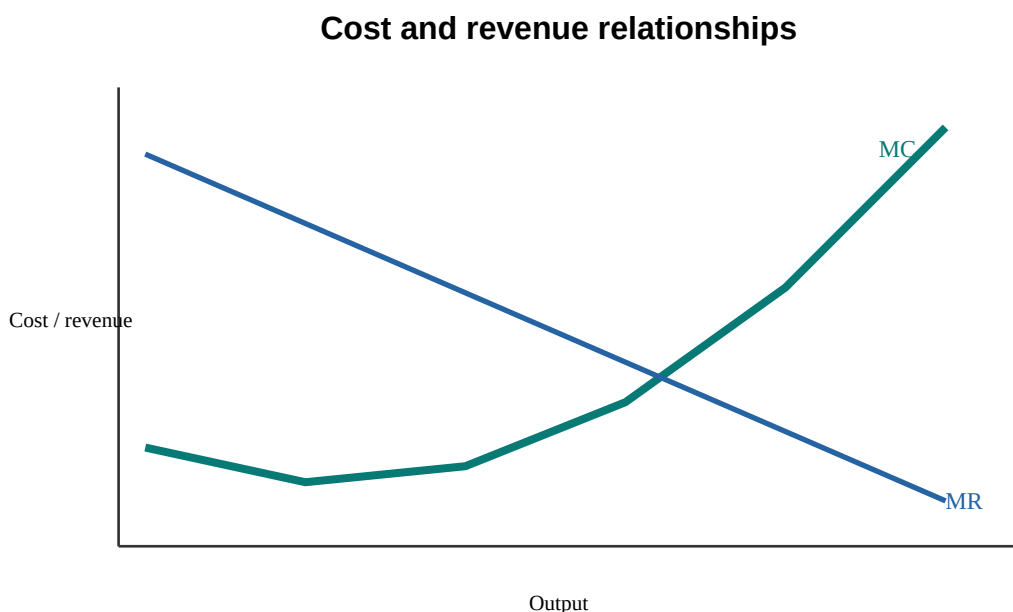
## 条件与评价 | Conditions and evaluation

Natural monopoly

中重复基础设施成本高，受监管单一企业可能比许多小企业有效；其他市场则竞争更可能保护消费者。

评价不是另起一个无关观点，而是改变时间、幅度、弹性、资源条件或利益相关者后，检查原结论是否仍成立。

## 图表推演 | Diagram analysis



考试图表必须标出坐标轴、曲线名称、新旧均衡和变化方向。文字解释需说明原因、曲线移动以及价格或数量结果。

## 英文精讲 | English explanation

Economies of scale lower long-run average cost as output expands. Diseconomies may arise from coordination problems.

## IGCSE 大考口径 | Cambridge exam focus

大考口径：准确定义；紧扣题干应用；解释完整因果链；评价结论说明条件。

## 第 18 课 | 2.5 从成本表到企业决策

所属大章：微观经济决策者 | Microeconomic decision makers

### 知识点精讲 | Core knowledge

把题中的每一列先翻译：TC 是总成本； $AC = TC/Q$ ； $TR = P \times Q$ ； $profit = TR - TC$ 。若固定成本不变，产量上升时 AC 可能下降，因为固定成本被更多单位分摊；这不代表每一项成本下降。

例子：厂房租金每月 10 000，生产 1 000 件时每件固定成本 10；生产 2 000 件时每件固定成本 5。若增加产量需要加班和更昂贵的原料，variable cost 仍可能上升。因此“规模扩大一定更便宜”错误。

写工会题时分两条线：工资提高会提升会员收入；企业劳动力成本提高可能减少雇佣、提高价格或投资自动化。结论取决于 labour demand elasticity、生产率和企业利润率。

### 深度展开与真题化应用 | In-depth development

#### 读表顺序决定是否出错

先识别 output，再找固定成本是否在各产量相同；计算  $VC = TC - FC$ ；再计算  $AC = TC / \text{output}$ 、 $TR = \text{price} \times \text{quantity}$  和  $profit = TR - TC$ 。零产量时 TC 通常等于 FC，但 AC 无法用 0 作除数。不要因为 AC 下降就写 TC 下降，也不要把 revenue 当 profit。

#### 数据变化要解释原因

AC 下降可能来自固定成本分摊、specialisation 或更高 machinery utilisation；随后上升可能来自 overtime、拥堵和协调问题。利润最大并不一定发生在 output 最大处，因为额外产量也带来额外成本。IGCSE 通常不要求边际分析，但必须比较收益与成本。

#### English development

Show each calculation clearly and include the correct unit. When interpreting a table, state both the change and the economic reason. For example, average cost falls from \$12 to \$9 because total cost rises less proportionately than output.

#### 真题化知识应用

若  $Q=50$ 、 $FC=200$ 、 $VC=300$ 、 $\text{price}=14$ ，则  $TC=500$ 、 $AC=10$ 、 $TR=700$ 、 $profit=200$ 。若扩大到  $Q=80$  后  $AC=11$ ，即使 TR 增加，也需重新计算利润才能判断扩张是否有利。

### 逐题精讲与评价拓展 | Exam workshop

#### 进一步拆解：这一课真正考什么

成本表按 Q、FC、VC、TC、AC、TR、profit

顺序处理。先确认单位和时间，再写公式与代入。解释数据时要说“由多少变到多少”和为什么，不只写 increase/decrease。

#### 完整因果链 | Cause-and-effect chain

output expands  $\rightarrow$  fixed cost spread over more units  $\rightarrow$  AFC falls  $\rightarrow$  AC may fall  $\rightarrow$  lower break-even price  $\rightarrow$  competitiveness improves；但 overtime/coordination may later raise variable cost per unit。

写分析题时把每个箭头改成 because 或 therefore 连接的完整句子；如果中间一步无法解释，说明链条仍有跳跃。

#### 数字与情境示范 | Numerical or contextual example

$Q=0$  时  $FC=240$ 、 $TC=240$ 。 $Q=40$  时  $TC=600$ ，所以  $VC=360$ 、 $AC=15$ 。 $\text{price}=18$  时  $TR=720$ 、 $profit=120$ 。若把 revenue 720 当 profit，会忽略全部成本。

## English guided explanation

A table may contain enough information to derive missing values. Show the formula, substitution and unit. A fall in average cost means cost per unit falls; it does not mean the firm spends less in total.

## Cambridge-style multiple-choice application

Question. At output 50, total cost is \$900 and variable cost is \$650. What is fixed cost?

Answer and reasoning. \$250. Fixed cost equals total cost minus variable cost.

选择题不仅要找到正确项，还要说明其余方向为什么与定义、公式或因果关系不符。

## Cambridge-style structured application

Question. Analyse why a firm may continue producing even when total profit falls temporarily.

Model development. It may expect demand to recover, wish to retain skilled workers or avoid high closure and restart costs. If revenue covers variable cost and contributes to fixed cost, short-run production can reduce losses. In the long run, persistent revenue below total cost is unsustainable.

## 高频混淆点 | Common misconception

零产量时 AC 不可计算，因为不能除以零；profit 可能为负，称 loss。

## 条件与评价 | Conditions and evaluation

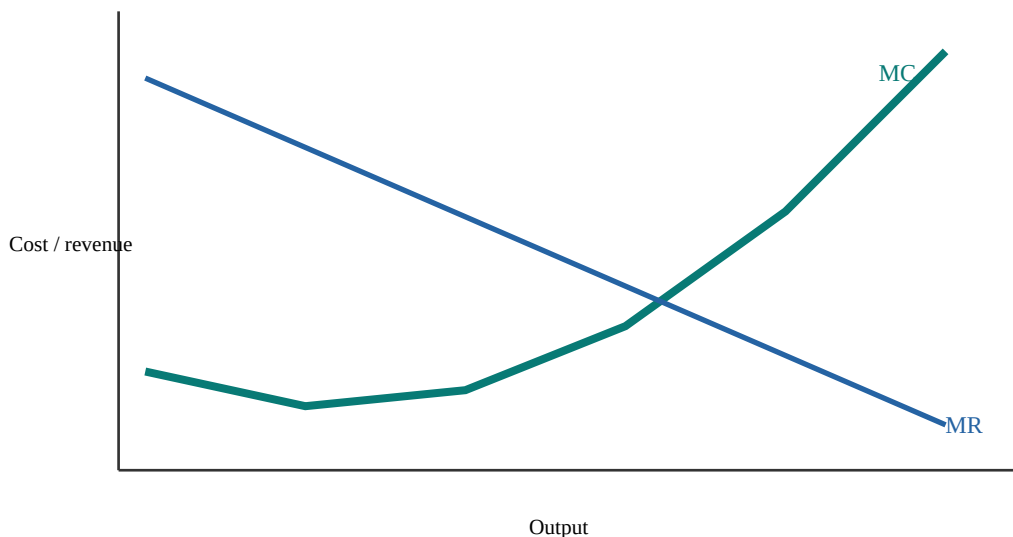
企业决策还受 cash flow、capacity、competitor behaviour、quality

与长期战略影响，不能只由一行平均成本决定。

评价不是另起一个无关观点，而是改变时间、幅度、弹性、资源条件或利益相关者后，检查原结论是否仍成立。

## 图表推演 | Diagram analysis

### Cost and revenue relationships



考试图表必须标出坐标轴、曲线名称、新旧均衡和变化方向。文字解释需说明原因、曲线移动以及价格或数量结果。

## 英文精讲 | English explanation

Total cost equals fixed cost plus variable cost; average cost is total cost divided by output. Profit equals total revenue minus total cost.



## IGCSE 大考口径 | Cambridge exam focus

大考口径：准确定义；紧扣题干应用；解释完整因果链；评价结论说明条件。

## 第 19 课 | 2.1 宏观目标与指标

所属大章: 政府与宏观经济 | Government and the macroeconomy

### 知识点精讲 | Core knowledge

常见目标是经济增长、低失业、低且稳定通胀、国际收支平衡与更公平的收入分配。它们会冲突: 刺激总需求或许降低周期性失业, 但也可能提高通胀和进口, 恶化经常账户。

### 深度展开与真题化应用 | In-depth development

#### 五类目标与测量口径

主要目标包括 economic growth、low unemployment、price stability、redistribution of income 和 balance of payments stability。Growth 用 real GDP 的变化衡量; 通胀用 consumer price index 的持续增长; 失业率=unemployed/labour force  $\times 100$ ; 收入分配可结合 Lorenz curve; 国际交易重点看 current account。

#### 目标之间既有协同也有冲突

衰退时提高 AD 可能同时增加 output 和 employment; 接近满负荷时继续刺激 AD 更可能推高 inflation。快速增长可能增加 imports、恶化 current account, 也可能通过生产率和出口改善它。重新分配收入可能提高低收入者消费, 却可能影响工作、储蓄和投资激励。

#### English development

Macroeconomic performance must be judged using several indicators. A lower unemployment rate may raise output and tax revenue, but labour shortages can increase wage pressure. Stable prices do not mean that every price is unchanged; they mean the general price level rises slowly and predictably.

#### 真题化知识应用

数据题要求比较国家时, 不只抄数值: 指出 trend、difference、relationship 和 exception。若 life expectancy 较高的国家通常 net immigration 较高, 要引用至少两组数据, 再指出反例并解释其他影响因素。

### 逐题精讲与评价拓展 | Exam workshop

#### 进一步拆解: 这一课真正考什么

宏观评价需要同时看 real GDP growth、unemployment、inflation、income distribution 和 current account。每项指标有测量误差, 且政策改善一个目标可能恶化另一个。

#### 完整因果链 | Cause-and-effect chain

AD rises when spare capacity exists  $\rightarrow$  firms increase output  $\rightarrow$  derived demand for labour rises  $\rightarrow$  unemployment falls  $\rightarrow$  household income and tax revenue rise; 接近满产时则 inflation pressure 更大。写分析题时把每个箭头改成 because 或 therefore 连接的完整句子; 如果中间一步无法解释, 说明链条仍有跳跃。

#### 数字与情境示范 | Numerical or contextual example

失业人数 0.6m、labour force 12m, unemployment rate=5%。若失业人数不变但更多人进入 labour force, 失业率可能变化, 因此分母也要检查。

#### English guided explanation

Macroeconomic objectives are related but not identical. Economic growth is measured using real output; price stability refers to low and predictable inflation; full employment does not mean zero unemployment because frictional unemployment normally exists.

## Cambridge-style multiple-choice application

Question. Which policy outcome shows a conflict: lower unemployment accompanied by higher demand-pull inflation, or higher productivity with lower costs?

Answer and reasoning. Lower unemployment with higher demand-pull inflation, because improvement in one objective is associated with deterioration in another.

选择题不仅要找到正确项，还要说明其余方向为什么与定义、公式或因果关系不符。

## Cambridge-style structured application

Question. Discuss whether low unemployment is always beneficial to an economy.

Model development. It raises output, income, tax revenue and may reduce benefit spending and poverty. However, severe labour shortages can increase wage costs and inflation, reduce competitiveness and make recruitment difficult. Some job movement is useful. The ideal rate depends on productivity and labour mobility.

## 高频混淆点 | Common misconception

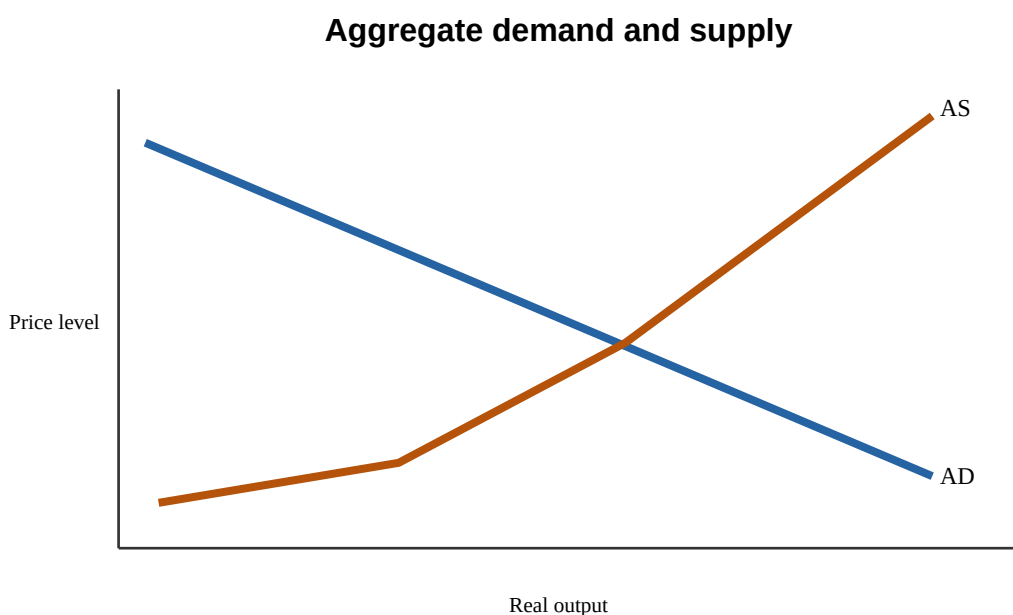
Price stability 不等于 price level 为零或每种商品价格不变；通胀下降也不等于总体价格下降。

## 条件与评价 | Conditions and evaluation

比较国家时须使用相同年份、定义与单位，并注意 informal sector、underemployment 和地区差异。

评价不是另起一个无关观点，而是改变时间、幅度、弹性、资源条件或利益相关者后，检查原结论是否仍成立。

## 图表推演 | Diagram analysis



考试图表必须标出坐标轴、曲线名称、新旧均衡和变化方向。文字解释需说明原因、曲线移动以及价格或数量结果。

## 英文精讲 | English explanation

Macroeconomic objectives include growth, employment, price stability, redistribution and balance-of-payments stability.

## IGCSE 大考口径 | Cambridge exam focus

大考口径：准确定义；紧扣题干应用；解释完整因果链；评价结论说明条件。

## 第 20 课 | 2.2 经济增长、失业与通胀

所属大章: 政府与宏观经济 | Government and the macroeconomy

### 知识点精讲 | Core knowledge

Economic growth 是实际 GDP

增加; 它可提高收入与税收, 但若只依赖资源开采, 可能污染并加剧不平等。失业分为 frictional、structural、cyclical 和 seasonal。通胀是总体价格水平持续上升, 不是单一商品涨价; demand-pull inflation 来自总需求增长过快, cost-push inflation 来自工资、能源或进口成本上升。

### 深度展开与真题化应用 | In-depth development

#### 三个概念必须分开

Economic growth 是 real GDP 增加; recession 通常是 real output 下降。Inflation 是总体价格水平持续上升, 不是一个商品涨价。Unemployment 指愿意且能够工作、正在寻找工作却没有工作的人。Frictional、structural、cyclical、seasonal unemployment 的原因和政策不同。

#### 原因与影响的双向链条

Demand-pull inflation 来自 AD 增长过快; cost-push inflation 来自工资、能源或进口成本上升。通胀降低固定收入者购买力并增加不确定性, 但温和通胀也可能伴随增长。失业降低 household income、tax revenue 和 output, 却减少工资与价格压力。增长可来自资源增加、productivity、investment 和技术进步。

#### English development

Nominal GDP may rise only because prices rise, so real GDP is required to measure growth. Structural unemployment occurs when workers' skills or locations do not match available jobs. Policies should address the cause: demand management may reduce cyclical unemployment, whereas retraining is more suitable for structural unemployment.

#### 真题化知识应用

若 nominal GDP 增 7%、price level 增 5%, real output 大致只增约 2%, 不能说生活水平增 7%。解释 cost-push inflation: oil price rises → firms' costs increase → aggregate supply falls → general price level rises and real output may fall。

### 逐题精讲与评价拓展 | Exam workshop

#### 进一步拆解: 这一课真正考什么

Growth、inflation、unemployment 三者的定义、原因和影响要分别建立, 再分析它们之间的联系。Nominal 与 real、rate of inflation 与 price level、unemployed 与 economically inactive 都是高频区别。

#### 完整因果链 | Cause-and-effect chain

oil price rises → firms' production costs rise → SRAS falls → price level rises and real output falls → cost-push inflation and cyclical unemployment may coexist。写分析题时把每个箭头改成 because 或 therefore 连接的完整句子; 如果中间一步无法解释, 说明链条仍有跳跃。

#### 数字与情境示范 | Numerical or contextual example

CPI 从 120 升到 126,  $\text{inflation} = (126 - 120) / 120 \times 100 = 5\%$ 。下一年 CPI 从 126 升到 129.15, 通胀 2.5%; 价格仍上涨, 只是上涨速度减慢。

## English guided explanation

Disinflation is a fall in the inflation rate while the price level continues to rise. Deflation is a sustained fall in the general price level. Structural unemployment reflects a mismatch of skills or location, while cyclical unemployment results from deficient aggregate demand.

## Cambridge-style multiple-choice application

Question. The inflation rate falls from 8% to 3%. Which statement is correct?

Answer and reasoning. The general price level is still rising, but more slowly. This is disinflation, not necessarily deflation. 选择题不仅要找到正确项，还要说明其余方向为什么与定义、公式或因果关系不符。

## Cambridge-style structured application

Question. Analyse two costs of high inflation for firms.

Model development. Uncertain future costs and prices make investment planning harder. If domestic inflation exceeds competitors', exports become less price competitive and imports more attractive. Firms may also face menu costs and pressure for higher wages.

## 高频混淆点 | Common misconception

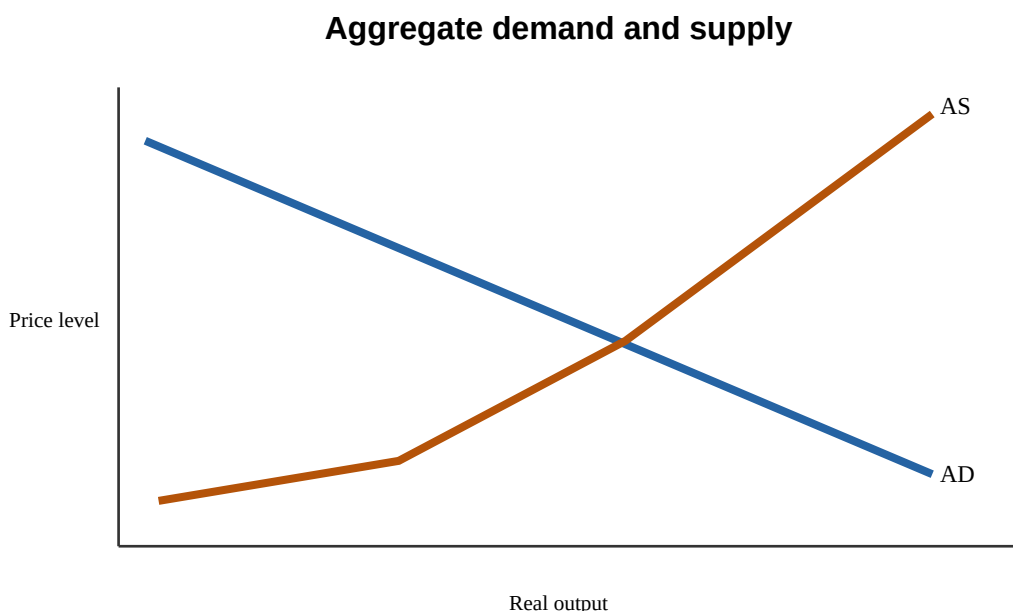
Real GDP per head may fall even when real GDP rises if population grows faster.

## 条件与评价 | Conditions and evaluation

适度需求增长可能同时降失业并带来温和通胀；长期结果取决于 productive capacity、expectations 和政策可信度。

评价不是另起一个无关观点，而是改变时间、幅度、弹性、资源条件或利益相关者后，检查原结论是否仍成立。

## 图表推演 | Diagram analysis



考试图表必须标出坐标轴、曲线名称、新旧均衡和变化方向。文字解释需说明原因、曲线移动以及价格或数量结果。

## 英文精讲 | English explanation

Growth is an increase in real output. Inflation is a sustained rise in the general price level. Unemployment has different causes.

## IGCSE 大考口径 | Cambridge exam focus

大考口径：准确定义；紧扣题干应用；解释完整因果链；评价结论说明条件。

## 第 21 课 | 2.3 政策工具 ( Policy tools )

所属大章: 政府与宏观经济 | Government and the macroeconomy

### 知识点精讲 | Core knowledge

Fiscal policy: 政府增加支出或减税以提高 AD; 代价可能是预算赤字与债务。

Monetary policy: 降低利率使借贷更便宜, 往往提高消费和投资, 但在信心低迷时效果弱。

Supply-side policy: 培训、教育、基础设施与竞争政策提高生产能力, 见效较慢但更能改善长期增长。

### 深度展开与真题化应用 | In-depth development

#### 三类政策作用点不同

Fiscal policy 改变 government spending 与 taxation; monetary policy 通过 interest rates、money/credit conditions 和 exchange rate 影响需求; supply-side policy 改善 education、infrastructure、competition、labour mobility 和 incentives, 提高 productive capacity。Expansionary 政策通常提高 AD, contractionary 政策降低 AD。

#### 政策传导不是一句“促进经济”

降息 → borrowing cheaper/saving less attractive → C 与 I 可能增加 → AD rises → firms raise output → employment rises。所得税下降 → disposable income rises → consumption rises。培训 → skills/productivity rise → unit cost falls → AS/productive potential rises。每条链都要说明可能条件: confidence、time lag、size、elasticity、debt 和 import leakage。

#### English development

Policy effectiveness depends on the cause of the problem and the state of the economy. Demand-side expansion may have a large output effect when spare capacity exists, but a larger inflationary effect near full employment. Supply-side policy often works slowly and has an opportunity cost.

#### 真题化知识应用

讨论政府是否应降税时, 正面发展 disposable income、consumption、AD、growth/employment; 反面写 budget deficit、inflation、imports、unequal benefits。结论说明经济是否衰退、税种、乘数大小和政府债务。

### 逐题精讲与评价拓展 | Exam workshop

#### 进一步拆解: 这一课真正考什么

Fiscal、monetary 与 supply-side policy

要按直接工具、传导渠道、目标和副作用区分。政策名称本身不得分, 必须写 households/firms 怎样因 incentive 改变行为。

#### 完整因果链 | Cause-and-effect chain

income tax cut → disposable income rises → consumption may rise → AD rises → firms increase output and employment → inflation/imports may also rise。training → skills/productivity rise → productive capacity increases, but after a time lag。写分析题时把每个箭头改成 because 或 therefore 连接的完整句子; 如果中间一步无法解释, 说明链条仍有跳跃。

#### 数字与情境示范 | Numerical or contextual example

政府支出增加\$100m不代表GDP恰好增加\$100m。Multiplier 可放大, imports、saving 与 taxes 会形成 leakage; 考试中没有数据时不要臆造精确倍数。

## English guided explanation

Fiscal policy changes government spending and taxation. Monetary policy changes interest rates or credit conditions. Supply-side policy aims to improve the quantity, quality or mobility of productive resources. Match the policy to the cause of the problem.

## Cambridge-style multiple-choice application

Question. Which policy is most directly designed to reduce structural unemployment: lower interest rates, retraining, higher government consumption, or a currency depreciation?

Answer and reasoning. Retraining, because it reduces the mismatch between workers' skills and available jobs. 选择题不仅要找到正确项，还要说明其余方向为什么与定义、公式或因果关系不符。

## Cambridge-style structured application

Question. Discuss whether higher government spending will increase economic growth.

Model development. Spending raises AD and may increase output when spare capacity exists. Infrastructure and education can also increase long-run productivity. However, spending may cause inflation, imports, taxation or debt, and inefficient projects have opportunity cost. The result depends on spending type and economic conditions.

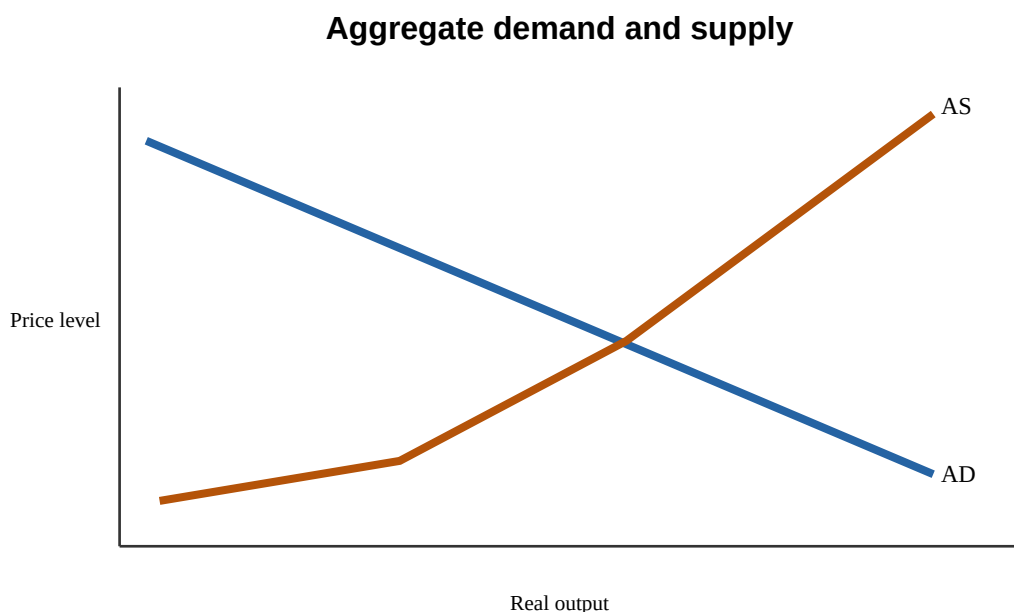
## 高频混淆点 | Common misconception

Supply-side policy 不一定表示政府减少供给；它旨在提高 economy's productive potential。

## 条件与评价 | Conditions and evaluation

评价至少加入 size、timing、confidence、spare capacity、opportunity cost 与 distribution。  
评价不是另起一个无关观点，而是改变时间、幅度、弹性、资源条件或利益相关者后，检查原结论是否仍成立。

## 图表推演 | Diagram analysis



考试图表必须标出坐标轴、曲线名称、新旧均衡和变化方向。文字解释需说明原因、曲线移动以及价格或数量结果。

## 英文精讲 | English explanation



Fiscal policy changes government spending and taxation; monetary policy affects interest rates; supply-side policy raises productive capacity.

## IGCSE 大考口径 | Cambridge exam focus

大考口径：准确定义；紧扣题干应用；解释完整因果链；评价结论说明条件。

## 第 22 课 | 2.4 国际收支 (Balance of payments)

所属大章: 政府与宏观经济 | Government and the macroeconomy

### 知识点精讲 | Core knowledge

经常账户记录货物、服务、收入和转移支付。赤字代表流出大于流入，不必然坏：若进口的是资本设备，可能提升未来生产力；但长期靠短期借款融资会有风险。分析时区分“赤字的原因”和“赤字本身”。

### 深度展开与真题化应用 | In-depth development

#### Current account 的组成与符号

Current account 包括 trade in goods、trade in services、primary income 和 secondary income。出口记录为 credit，进口为 debit。Surplus 表示流入大于流出，deficit 相反。它是流量，不是政府“欠款总额”，也不等同于财政预算赤字。

#### 赤字原因和影响需结合背景

进口高可能来自高收入、强汇率或本国产品竞争力低；也可能是进口 capital goods，为未来出口能力打基础。赤字可能增加外债或使汇率承压，但居民可获得更多商品和机器。改善方法包括 depreciation、提高 productivity、贸易限制或降低 AD，各自有副作用。

#### English development

A current-account deficit is not automatically harmful. It may reflect strong domestic demand or investment in imported capital. A depreciation can make exports cheaper and imports dearer, but quantities respond only over time and according to price elasticity.

#### 真题化知识应用

讨论 surplus 是否提高 living standards: 出口增长可提高 employment/income; 但 surplus 可能来自低 imports 和弱消费，也可能造成 inflation 或贸易摩擦。结论应结合 surplus 原因、收入分配和资金是否用于提高生产能力。

### 逐题精讲与评价拓展 | Exam workshop

#### 进一步拆解：这一课真正考什么

Current account 记录 goods、services、primary income 与 secondary income 的国际流量。区分 visible goods、invisible services、current-account balance、government budget 与 national debt。

#### 完整因果链 | Cause-and-effect chain

domestic income rises → consumers buy more imports → import expenditure rises → current-account balance may deteriorate; 若增长来自 export productivity，结果可能相反。写分析题时把每个箭头改成 because 或 therefore 连接的完整句子；如果中间一步无法解释，说明链条仍有跳跃。

#### 数字与情境示范 | Numerical or contextual example

Goods exports 80、imports 100; services exports 40、imports 25; net primary income -5、transfers +2。Current balance =  $(80 - 100) + (40 - 25) - 5 + 2 = -8$ ，为 deficit 8。

#### English guided explanation

A current-account deficit means payments out exceed receipts in the recorded categories. It is not identical to government debt. A surplus may reflect strong exports, weak domestic demand or restricted imports, so its welfare effect is ambiguous.

## Cambridge-style multiple-choice application

Question. Which transaction is a credit item: a resident buys an imported car, a foreign tourist buys a hotel service domestically, or government pays foreign interest?

Answer and reasoning. The foreign tourist's purchase, because it is an export of services and brings foreign currency receipts. 选择题不仅要找到正确项，还要说明其余方向为什么与定义、公式或因果关系不符。

## Cambridge-style structured application

Question. Discuss whether a current-account deficit is harmful.

Model development. It may weaken the currency, increase borrowing and indicate low competitiveness. Yet it may finance imported capital goods, reflect strong growth and provide consumers with choice. Harm depends on size, persistence, financing and whether imports improve future productivity.

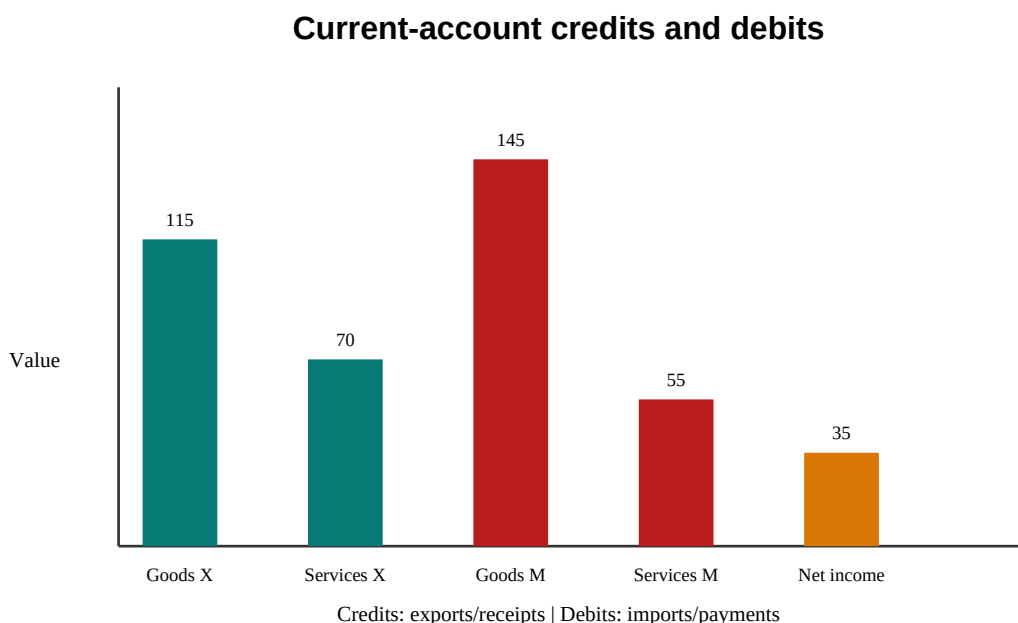
## 高频混淆点 | Common misconception

Trade balance 只含 goods (有些口径含 goods and services 时必须看题目)；current account 范围更广。

## 条件与评价 | Conditions and evaluation

判断 balance 不能脱离 GDP 比例、持续时间、foreign reserves、exchange-rate regime 与进口结构。  
评价不是另起一个无关观点，而是改变时间、幅度、弹性、资源条件或利益相关者后，检查原结论是否仍成立。

## 图表推演 | Diagram analysis



考试图表必须标出坐标轴、曲线名称、新旧均衡和变化方向。文字解释需说明原因、曲线移动以及价格或数量结果。

## 英文精讲 | English explanation

The current account records trade in goods and services, primary income and secondary income.

## IGCSE 大考口径 | Cambridge exam focus

大考口径：准确定义；紧扣题干应用；解释完整因果链；评价结论说明条件。

## 第 23 课 | 2.5 政策传导链的写法

所属大章：政府与宏观经济 | Government and the macroeconomy

### 知识点精讲 | Core knowledge

不要只写“降低利率促进经济”。拆开写：interest rate falls → monthly loan repayments fall / borrowing becomes cheaper → households consume more and firms invest more → AD rises → output and employment may rise.

然后评价：如果家庭担心失业，可能仍储蓄；若经济接近满产，效果更可能是通胀而不是实际产出。

财政政策同理：government spending on roads → firms receive orders → workers' incomes rise → further consumption occurs. 但政府融资可能增加债务；道路项目也有长期 supply-side benefit。把短期 AD 效果与长期 LRAS 效果分开，答案会更成熟。

### 深度展开与真题化应用 | In-depth development

每个箭头都要有 because

高分 analysis 不是列出变量，而是解释关系。例如 “interest rate ↓ → investment ↑ → AD ↑ → GDP ↑” 仍不够完整；应写借款成本下降提高预期净收益，使更多投资项目有利可图，capital spending 增加，AD 上升，企业面对更高销售而增加产量与就业。

在链条末端加入限制条件

若企业缺乏 confidence，即使利率下降也可能不借款；若经济接近 full employment，AD 增长更多转化为 inflation；若进口倾向高，本国 output 增幅较小。Supply-side policy 需要教育和基建完成后才见效，time lag 往往更长。

English development

A developed chain identifies the initial policy, the incentive it changes, the behaviour of households or firms, and the final macroeconomic outcome. Evaluation then changes one assumption: spare capacity, confidence, elasticity, time or policy size.

真题化知识应用

六分 analyse 题通常需要两条独立且发展的链，而不是六个短词。写完后检查是否回答题干主体，例如“对 firms 的影响”应落到 costs、revenue、profit、investment 或 employment，而非只停在 GDP。

### 逐题精讲与评价拓展 | Exam workshop

进一步拆解：这一课真正考什么

Analyse 题的核心是中间机制。每条链应包含 initial change、incentive、agent response、market/macro variable 和 final outcome。箭头之间都能回答 because，才算 developed analysis。

完整因果链 | Cause-and-effect chain

interest rate falls → monthly loan cost falls → households may borrow for durable goods → consumption rises → AD rises → firms receive more orders → output and employment rise, if spare capacity exists. 写分析题时把每个箭头改成 because 或 therefore 连接的完整句子；如果中间一步无法解释，说明链条仍有跳跃。

数字与情境示范 | Numerical or contextual example

不要用没有题干依据的精确数字，但可以用方向检查。若税后收入由\$1 000升到\$1 100而家庭只消费新增\$60，剩余\$40 saving 是 leakage，AD 增量小于收入增量。

## English guided explanation

A causal chain should not jump from policy to the final objective. Explain the transmission through incentives and behaviour. Separate a direct effect from secondary effects and avoid repeating the same link in different words.

## Cambridge-style multiple-choice application

Question. Which is the missing link: lower interest rate → \_\_\_\_ → higher investment → higher AD?

Answer and reasoning. Borrowing becomes cheaper or the expected net return on investment rises.

选择题不仅要找到正确项，还要说明其余方向为什么与定义、公式或因果关系不符。

## Cambridge-style structured application

Question. Analyse how a depreciation may reduce unemployment.

Model development. Exports become cheaper in foreign currency and imports become dearer domestically. If demand responds, export sales rise and spending switches to domestic products. Firms increase output, raising derived demand for labour and reducing cyclical unemployment.

## 高频混淆点 | Common misconception

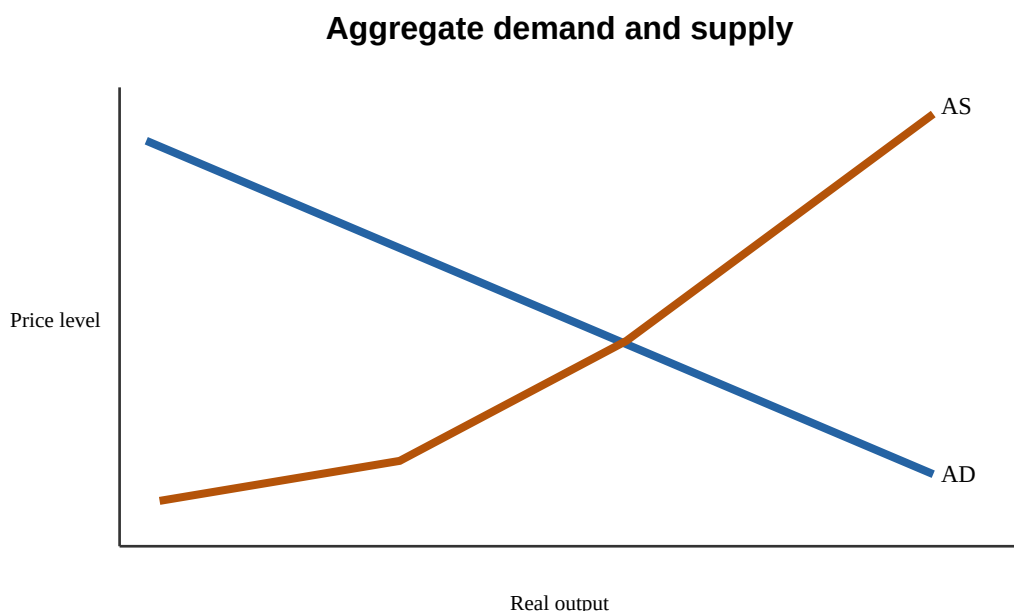
“政策促进经济”不是因果链。必须写谁的成本、收入或激励改变，以及行为如何改变。

## 条件与评价 | Conditions and evaluation

每条链末端加入一个与题目有关的条件即可，例如 PED、confidence、time lag 或 spare capacity，不要罗列无关限制。

评价不是另起一个无关观点，而是改变时间、幅度、弹性、资源条件或利益相关者后，检查原结论是否仍成立。

## 图表推演 | Diagram analysis



考试图表必须标出坐标轴、曲线名称、新旧均衡和变化方向。文字解释需说明原因、曲线移动以及价格或数量结果。

## 英文精讲 | English explanation

Fiscal policy changes government spending and taxation; monetary policy affects interest rates; supply-side policy raises productive capacity.

## IGCSE 大考口径 | Cambridge exam focus

大考口径：准确定义；紧扣题干应用；解释完整因果链；评价结论说明条件。

## 第 24 课 | 2.1 增长不等于发展 (Growth versus development)

所属篇章: 经济发展 | Economic development

### 知识点精讲 | Core knowledge

增长是实际 GDP 上升; development (发展) 是生活质量和选择能力的更广泛改善。一个国家可因石油出口 GDP 增长, 却仍有贫困、教育不足、污染和地区差距。因此答案中把“人均实际 GDP”与健康、教育、住房、自由、安全、环境一起写。

### 深度展开与真题化应用 | In-depth development

#### 两个概念的范围不同

Growth 是 real GDP 增加, development 是生活质量的广泛改善, 包括 income、health、education、housing、environment 和 freedom。增长可增加税基和就业, 为发展提供资源; 但若收益集中在少数人、污染严重或人口增长更快, 平均居民未必受益。

#### 因果关系不是自动发生

资源出口提高 GDP 后, 政府若把收入用于 clean water、schools 和 clinics, human capital 与 life expectancy 可提高; 若收入外流、腐败或只进入资本密集行业, poverty 可能仍高。Real GDP per head 比总 GDP 更接近平均物质生活, 但仍忽略 distribution 和非货币因素。

#### English development

Economic growth is an increase in real national output. Economic development is a multidimensional improvement in welfare. Growth can finance development, but its effect depends on population growth, income distribution, environmental costs and government policy.

#### 真题化知识应用

“Discuss whether growth always raises living standards” 应包含 income/employment/tax revenue 的正面链, 也要发展 inflation、inequality、long hours、resource depletion 和 pollution。结论依据 growth 的来源、持续性和收益分配。

### 逐题精讲与评价拓展 | Exam workshop

#### 进一步拆解: 这一课真正考什么

Growth 是 real output 的数量变化; development 是生活质量的多维变化。二者可相互促进, 但不具自动因果。分析需经过 real GDP per head、distribution、public services 和 environment。

#### 完整因果链 | Cause-and-effect chain

export-led growth → employment/income/tax revenue rise → government may fund health and education → human capital and life expectancy improve → development; 若利润外流或污染严重, 链条可能中断。写分析题时把每个箭头改成 because 或 therefore 连接的完整句子; 如果中间一步无法解释, 说明链条仍有跳跃。

#### 数字与情境示范 | Numerical or contextual example

Real GDP 增6%、人口增4%, real GDP per head 大致增约2%, 不是6%。若最高10%获得大部分新增收入, median household 的生活水平可能几乎不变。

#### English guided explanation

Growth expands the amount an economy produces. Development concerns what the output enables people to achieve. Per-head measures, distribution and non-material indicators are therefore essential when assessing welfare.

### Cambridge-style multiple-choice application

Question. Which most clearly indicates development rather than growth alone: higher real GDP, longer life expectancy, higher price level, or more currency in circulation?

Answer and reasoning. Longer life expectancy, because it directly reflects a health and welfare outcome. 选择题不仅要找到正确项，还要说明其余方向为什么与定义、公式或因果关系不符。

### Cambridge-style structured application

Question. Discuss whether rapid economic growth improves living standards.

Model development. Growth may create jobs, raise incomes and tax revenue for services. It can finance infrastructure. However, population may grow faster, income may be unequal, work hours may rise and pollution or resource depletion may worsen. The result depends on inclusiveness and sustainability.

### 高频混淆点 | Common misconception

Nominal GDP 增加可能只是 inflation; total GDP 增加也可能被 population growth 抵消。

### 条件与评价 | Conditions and evaluation

结论最好区分 short run material gain 与 long-run sustainable development。

评价不是另起一个无关观点，而是改变时间、幅度、弹性、资源条件或利益相关者后，检查原结论是否仍成立。

## 英文精讲 | English explanation

Economic development is broader than growth and includes improvements in health, education and living standards.

## IGCSE 大考口径 | Cambridge exam focus

大考口径：准确定义；紧扣题干应用；解释完整因果链；评价结论说明条件。



## 第 25 课 | 2.2 衡量生活水平 ( Living standards )

所属大章: 经济发展 | Economic development

### 知识点精讲 | Core knowledge

GDP per head 适合粗略比较物质产出, 但忽略收入分配、非市场生产、工作时长与环境。HDI 结合寿命、教育和收入, 覆盖更广; 仍是平均数, 会掩盖性别、城乡与地区不平等。比较国家时还要注意汇率、购买力和人口结构。

### 深度展开与真题化应用 | In-depth development

#### 单一指标与综合指标

Real GDP per head=real GDP/population, 适合比较平均物质产出, 但跨国比较还要考虑 exchange rate、purchasing power、informal economy 和 data accuracy。HDI 结合 income、education 和 life expectancy, 范围更广, 却仍是平均值, 不能完整反映 inequality、security 或 environment。

#### 读数据要比较、联系并找例外

一国人均收入高而 life expectancy

低, 可能说明不平等、医疗不足或污染; 另一国收入较低但教育健康良好, 可能源于公共服务效率。Correlation 不等于 causation, 还要考虑其他变量。比较时应使用同一年、相同单位和 real/per-head 数据。

#### English development

An indicator is useful only if its measurement and limitation are understood. Real GDP per head allows for inflation and population, while HDI broadens the comparison. Neither reveals how income is distributed within the country.

#### 真题化知识应用

数据题先写总体关系, 如 higher income is generally associated with longer life expectancy; 再引用两组具体数值支持; 找一个 exception; 最后解释 healthcare、diet、sanitation 或 inequality。不要逐行抄表。

### 逐题精讲与评价拓展 | Exam workshop

#### 进一步拆解: 这一课真正考什么

评价指标要问: 测量什么、是否 real、是否 per head、是否平均数、数据是否可比、忽略什么。Real GDP per head、HDI 与单项 social indicators 应互相补充。

#### 完整因果链 | Cause-and-effect chain

education improves → literacy and skills rise → productivity/employability rise → income may rise; health improves → fewer workdays lost → effective labour supply and wellbeing increase.

写分析题时把每个箭头改成 because 或 therefore 连接的完整句子; 如果中间一步无法解释, 说明链条仍有跳跃。

#### 数字与情境示范 | Numerical or contextual example

A国 real GDP 500bn、人口50m, 人均10 000; B国300bn、人口20m, 人均15 000。A总产出更高, 但B平均物质产出更高。还不能据此判断分配和健康。

#### English guided explanation

Real GDP per head is useful for average material output, while HDI includes income, education and life expectancy. Purchasing-power adjustments improve international comparison. No single index captures inequality, freedom, safety or environmental quality.

## Cambridge-style multiple-choice application

Question. Why is real GDP per head usually better than nominal GDP for comparing living standards over time?

Answer and reasoning. It adjusts for both price changes and population, though it remains an average. 选择题不仅要找到正确项，还要说明其余方向为什么与定义、公式或因果关系不符。

## Cambridge-style structured application

Question. Analyse two limitations of using GDP per head to compare countries.

Model development. Exchange rates may not reflect local purchasing power, so equal dollar incomes buy different amounts. Informal and unpaid activity may be recorded differently. The average also hides inequality, while pollution and leisure are omitted.

## 高频混淆点 | Common misconception

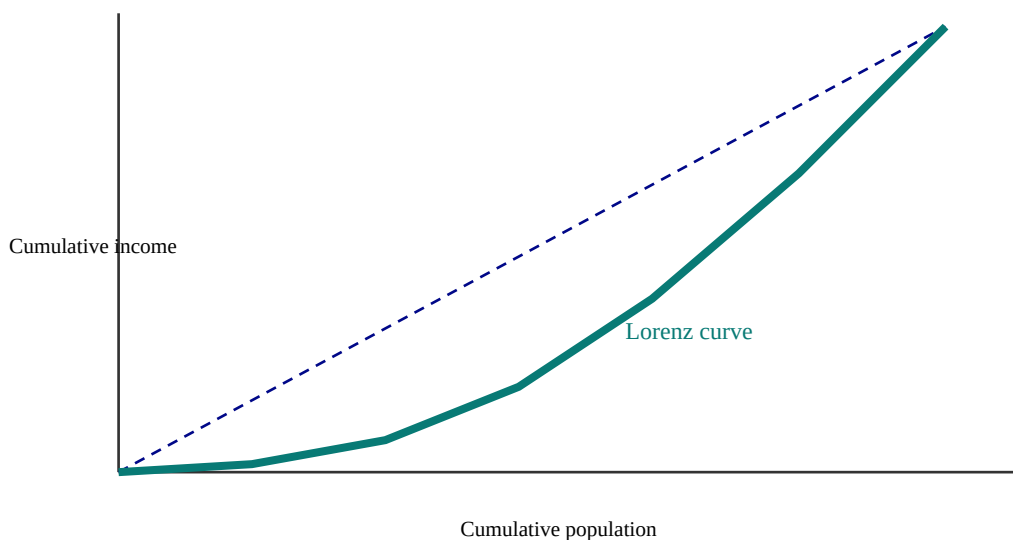
HDI 较高不表示每个分项都较高，也不代表国内人人处于相同水平。

## 条件与评价 | Conditions and evaluation

选择指标取决于问题：物质消费可看 real income，健康看 life expectancy，综合比较用多指标和分布数据。评价不是另起一个无关观点，而是改变时间、幅度、弹性、资源条件或利益相关者后，检查原结论是否仍成立。

## 图表推演 | Diagram analysis

### Lorenz curve and inequality



考试图表必须标出坐标轴、曲线名称、新旧均衡和变化方向。文字解释需说明原因、曲线移动以及价格或数量结果。

## 英文精讲 | English explanation

Real GDP per head helps compare material living standards, while HDI adds health and education.

## IGCSE 大考口径 | Cambridge exam focus

大考口径：准确定义；紧扣题干应用；解释完整因果链；评价结论说明条件。

## 第 26 课 | 2.3 贫困、人口与发展障碍

所属篇章: 经济发展 | Economic development

### 知识点精讲 | Core knowledge

绝对贫困指基本需要难以满足; 相对贫困强调与社会正常生活水平的差距。人口快速增长可扩大劳动年龄人口和市场, 也可能压低人均资本、增加学校医疗负担与失业。低储蓄、基础设施不足、政治不稳定、初级产品依赖和债务均可形成发展陷阱。

### 深度展开与真题化应用 | In-depth development

#### 贫困的两种口径

Absolute poverty 指收入不足以满足 food、shelter 等基本需要; relative poverty 指收入显著低于社会通常水平。低收入→营养教育不足→productivity 低→工资低, 形成 poverty cycle。税收转移、教育医疗和就业增长可能打破循环, 但政策有财政成本和激励问题。

#### 人口数量之外还有结构

Birth rate、death rate、life expectancy 和 net migration 改变人口规模; age structure 与 dependency ratio 影响 labour supply、tax revenue 和公共支出。年轻人口可能带来未来劳动力优势, 也需要当前教育投入; 老龄化增加养老金和医疗压力, 但也可能扩大养老服务需求。

#### English development

Population growth can increase the labour force and market size, but it may reduce living standards if output, housing and public services do not keep pace. Migration affects both the size and skill composition of the labour force.

#### 真题化知识应用

讨论高 birth rate 时, 不只写“人口多”: 短期 children dependency rises→education/health spending rises; 长期 labour force may grow→productive capacity and tax revenue rise。结论取决于 job creation、human-capital investment 和资源承载力。

### 逐题精讲与评价拓展 | Exam workshop

#### 进一步拆解: 这一课真正考什么

贫困部分区分 absolute 与 relative; 人口部分区分 total population、growth rate、age structure、labour force 和 dependency。二者通过 education、health、employment 与 public finance 相互作用。

#### 完整因果链 | Cause-and-effect chain

low income → poor nutrition/education → low productivity → low wages → limited saving/investment → poverty persists。政策可在 education、health、credit 或 job creation 处打断循环。写分析题时把每个箭头改成 because 或 therefore 连接的完整句子; 如果中间一步无法解释, 说明链条仍有跳跃。

#### 数字与情境示范 | Numerical or contextual example

工作年龄人口600万, 儿童250万, 老人150万, dependency ratio=(250+150)/600×100=66.7%。这不表示每100名工人实际供养66.7人, 因为工作年龄者并非全就业。

#### English guided explanation

Absolute poverty concerns inability to meet basic needs. Relative poverty compares a household with prevailing living standards. A high dependency ratio may increase pressure on workers and public services,

but dependants can also contribute unpaid work and future labour.

### Cambridge-style multiple-choice application

Question. Which change directly increases an ageing dependency ratio: higher birth rate, longer life expectancy with low fertility, net immigration of young workers, or later retirement?

Answer and reasoning. Longer life expectancy with low fertility, because the elderly share rises relative to the working-age population. 选择题不仅要找到正确项，还要说明其余方向为什么与定义、公式或因果关系不符。

### Cambridge-style structured application

Question. Discuss whether population growth benefits a developing economy.

Model development. It can expand labour supply, demand and tax base, encouraging investment. A young population can become a demographic dividend. But education, housing and jobs must expand; otherwise unemployment, congestion and pressure on resources may rise.

### 高频混淆点 | Common misconception

低 birth rate 不必然使人口立即下降；migration、death rate 和 population momentum 也会影响。

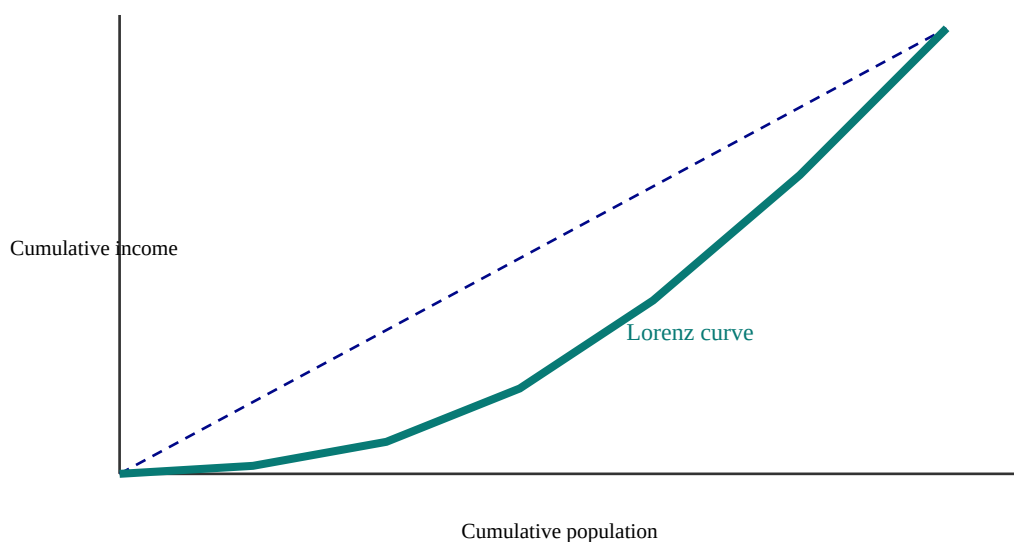
### 条件与评价 | Conditions and evaluation

人口结果取决于年龄结构、人力资本、女性就业、工作机会和基础设施，而非只看人数。

评价不是另起一个无关观点，而是改变时间、幅度、弹性、资源条件或利益相关者后，检查原结论是否仍成立。

## 图表推演 | Diagram analysis

### Lorenz curve and inequality



考试图表必须标出坐标轴、曲线名称、新旧均衡和变化方向。文字解释需说明原因、曲线移动以及价格或数量结果。

## 英文精讲 | English explanation

Absolute poverty concerns basic needs; relative poverty compares income with social norms.

## IGCSE 大考口径 | Cambridge exam focus

大考口径：准确定义；紧扣题干应用；解释完整因果链；评价结论说明条件。

## 第 27 课 | 2.4 发展策略的评价

所属篇章: 经济发展 | Economic development

### 知识点精讲 | Core knowledge

教育、医疗、基础设施、微型信贷、贸易开放、FDI 与援助都可能有用。例: 修港口可降低出口运输成本, 但若政府腐败、维护资金不足, 项目效益会很低。评价要问: 谁受益? 资金来自哪里? 短期与长期效果? 环境是否可持续?

### 深度展开与真题化应用 | In-depth development

#### 政策通过生产率改变发展

Education raises skills and occupational mobility; healthcare reduces illness and improves effective labour supply; infrastructure lowers transport and communication costs; microfinance may support enterprise; property rights and stable institutions encourage investment. 短期补贴能缓解贫困, 但长期发展通常要求生产能力与机会增加。

#### 评价预算、时滞与受益者

同一笔预算投向高速公路会产生 opportunity cost, 例如少建学校。大型项目可能偏向城市, 农村贫困者未必直接受益。教育见效慢但外部收益广; 现金转移见效快却需持续财政来源。Foreign aid 和 FDI 可带来资金技术, 也可能有条件、利润外流和 dependency。

#### English development

Development policy should be judged by effectiveness, cost, time period and distribution. Infrastructure may attract investment by lowering costs, but only if projects are well chosen and maintained. Education raises human capital, yet benefits take years to appear.

#### 真题化知识应用

八分 discuss

题可比较两种政策, 而不是堆砌四种名称。每种政策写机制、限制和受益者, 结论说明该国主要障碍若是 low skills, 则基础教育和职业培训可能比一般消费补贴更可持续。

### 逐题精讲与评价拓展 | Exam workshop

#### 进一步拆解: 这一课真正考什么

每项策略按 problem targeted、mechanism、time lag、cost、distribution 和 risk 评价。Education、health、infrastructure、redistribution、FDI 与 aid 的路径不同, 不应写成同一种“促进发展”。

#### 完整因果链 | Cause-and-effect chain

rural road investment → transport time/cost falls → farmers access larger markets → spoilage falls and revenue rises → saving/investment and rural living standards may improve。写分析题时把每个箭头改成 because 或 therefore 连接的完整句子; 如果中间一步无法解释, 说明链条仍有跳跃。

#### 数字与情境示范 | Numerical or contextual example

预算\$50m可建学校或发一年补贴。补贴立刻增加可支配收入; 学校需多年才转化为技能, 但收益可能持续更久。两者的 opportunity cost 与时间结构不同。

#### English guided explanation

A development strategy should address the binding constraint. Education is weak if graduates have no jobs; roads are less useful without maintenance; foreign investment brings fewer benefits when local firms and workers cannot form linkages.

### Cambridge-style multiple-choice application

Question. Which policy most directly raises human capital: a tariff, teacher training, currency appreciation, or a maximum price?

Answer and reasoning. Teacher training, because it can improve the quality of education and workers' future skills. 选择题不仅要找到正确项，还要说明其余方向为什么与定义、公式或因果关系不符。

### Cambridge-style structured application

Question. Discuss whether foreign aid is the best way to promote development.

Model development. Aid can finance health, education, disaster relief and infrastructure where domestic revenue is low. It may transfer expertise. However, tied aid, corruption, dependency and poor project selection reduce benefits. Trade access, governance or domestic taxation may be more sustainable.

### 高频混淆点 | Common misconception

投入资金不等于 outcome 改善；必须说明服务质量、覆盖人群与执行能力。

### 条件与评价 | Conditions and evaluation

“最佳”要求比较替代政策，并结合该国最主要障碍作判断。

评价不是另起一个无关观点，而是改变时间、幅度、弹性、资源条件或利益相关者后，检查原结论是否仍成立。

### 英文精讲 | English explanation

Education, healthcare and infrastructure may raise productivity, but involve opportunity cost and time lags.

### IGCSE 大考口径 | Cambridge exam focus

大考口径：准确定义；紧扣题干应用；解释完整因果链；评价结论说明条件。

## 第 28 课 | 2.5 指标题的比较框架

所属篇章: 经济发展 | Economic development

### 知识点精讲 | Core knowledge

看到“哪个国家生活水平更高”，不要只比较一个 GDP 数字。依次问：是否是 real GDP？是否 per head？汇率是否反映购买力？收入是否平均分配？教育、寿命、饮水和环境怎样？一个资源出口国的人均 GDP 高，仍可能有很大贫富差距。

HDI 的读法：它把 income、education 与 life expectancy 组合，所以能补充 GDP；但它是国家平均值，无法显示一个城市很富而农村很穷。可直接写：HDI is broader than GDP per head, although both figures can hide inequality within a country.

### 深度展开与真题化应用 | In-depth development

#### 指标比较的四层结构

第一层确认定义与单位；第二层描述 trend/difference；第三层用数据验证 relationship；第四层指出 exception 与 limitation。若比较 GDP，先确认 nominal or real、total or per head。若比较失业率，关注 labour-force 定义是否一致。

#### 平均值不代表每个人

GDP per head 上升可能与 median income 不同方向；HDI 改善也可能隐藏地区差异。数据只显示 correlation 时，不能直接写 A causes B。需要提出合理机制，并承认 reverse causation 或 third variable。

#### English development

Use comparative language: higher than, twice as high, increased by, increased from and percentage-point change. “Inflation rose by 2%” may be ambiguous; if it moves from 3% to 5%, it rises by two percentage points.

#### 真题化知识应用

若题目要求 analyse relationship，答题框架是 expected relationship → two pieces of evidence → economic explanation → exception → possible reason。这样的内容比把每个国家逐一复述更贴近 mark scheme。

### 逐题精讲与评价拓展 | Exam workshop

#### 进一步拆解：这一课真正考什么

数据题按 describe、compare、calculate、analyse

四种任务拆开。描述不解释；比较必须有双方；计算展示公式；分析要用数据证据支持机制并识别 exception。

#### 完整因果链 | Cause-and-effect chain

higher income may fund nutrition/sanitation/healthcare → mortality falls → life expectancy rises。但 inequality、conflict 或 pollution 可造成偏离总体关系的异常点。写分析题时把每个箭头改成 because 或 therefore 连接的完整句子；如果中间一步无法解释，说明链条仍有跳跃。

#### 数字与情境示范 | Numerical or contextual example

失业率从4%到6%是上升2 percentage

points，也是相对上升50%。题目若未指定，要用最清楚的表述，不能含混写“上升2%”。

#### English guided explanation



Use evidence selectively. A strong relationship answer states the overall pattern, cites two relevant comparisons, identifies an exception and offers an economic explanation. Correlation alone does not prove causation.

### Cambridge-style multiple-choice application

Question. Country X has twice the GDP of Y but four times the population. Which has higher GDP per head?

Answer and reasoning. Country Y. X has only half Y's GDP per head when both totals are divided by population. 选择题不仅要找到正确项，还要说明其余方向为什么与定义、公式或因果关系不符。

### Cambridge-style structured application

Question. Analyse a positive relationship between education spending and labour productivity shown in a table.

Model development. Higher spending may improve teacher quality and access, raising workers' skills. Skilled workers use capital more effectively and produce more per hour. Cite supporting countries and an exception; productivity also depends on technology, health and spending efficiency.

### 高频混淆点 | Common misconception

不要只抄最高值与最低值；relationship 题需要跨多组数据，exception 也必须有数值证据。

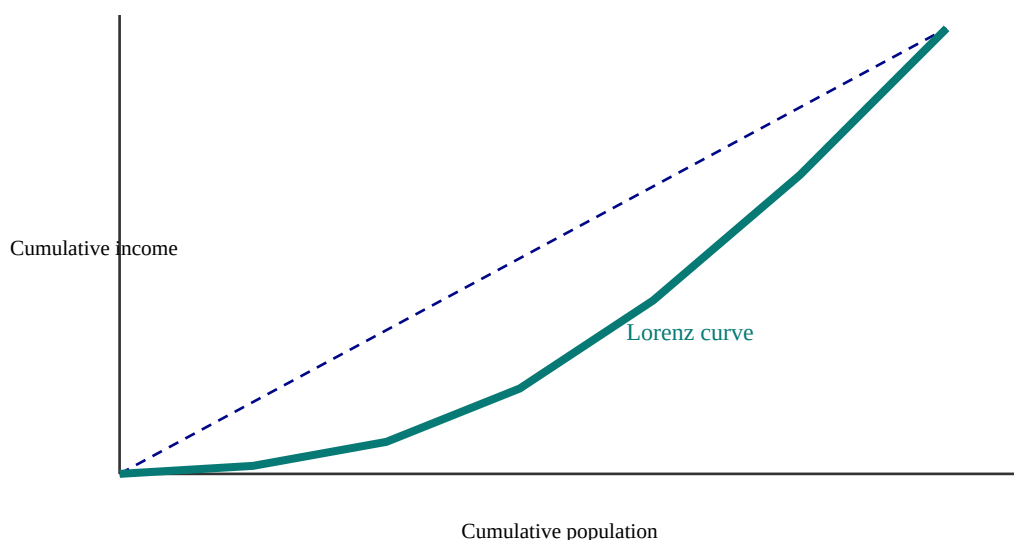
### 条件与评价 | Conditions and evaluation

数据年份、样本数、测量定义、滞后和第三变量限制因果结论。

评价不是另起一个无关观点，而是改变时间、幅度、弹性、资源条件或利益相关者后，检查原结论是否仍成立。

## 图表推演 | Diagram analysis

### Lorenz curve and inequality



考试图表必须标出坐标轴、曲线名称、新旧均衡和变化方向。文字解释需说明原因、曲线移动以及价格或数量结果。

### 英文精讲 | English explanation

Define the key term precisely, apply it to the case material and develop a complete economic explanation.

### IGCSE 大考口径 | Cambridge exam focus



大考口径：准确定义；紧扣题干应用；解释完整因果链；评价结论说明条件。

## 第 29 课 | 2.1 专业化与比较优势

所属大章: 国际贸易与全球化 | International trade and globalisation

### 知识点精讲 | Core knowledge

Specialisation 让个人、企业或国家集中在机会成本较低的生产活动。关键是 comparative advantage (比较优势), 不是绝对产量。即使 A 国两种产品都生产得更多, 只要 A 放弃制造品生产食品的代价较低, 它仍应专业化于食品, 双方通过贸易可消费到 PPC 以外的组合。

### 深度展开与真题化应用 | In-depth development

#### 分工、专业化与交换

Specialisation 让工人、企业、地区或国家集中于较少任务, 可提高 skill、速度和 machinery utilisation; 风险是 boredom、dependence 和需求变化。国家贸易方向应由 comparative advantage 决定, 即较低 opportunity cost, 而非只看 absolute output。

#### 机会成本数值法

A 每小时可生产 8 cloth 或 4 computers, 因此 1 computer 成本为 2 cloth; B 可生产 6 cloth 或 6 computers, 1 computer 成本为 1 cloth, 所以 B 在 computers 有比较优势。反过来, A 的 1 cloth 成本为 0.5 computer, 低于 B 的 1, 因此 A 在 cloth 有比较优势。互利交换比率必须位于两国机会成本之间。

#### English development

Absolute advantage means producing more with the same resources. Comparative advantage means producing at a lower opportunity cost. Even a country with an absolute advantage in both products can gain from trade if opportunity costs differ.

#### 真题化知识应用

表格题必须计算“双向单位成本”, 不要只看最大产量。若国际交换 1 computer 换 1.5 cloth, B 获得的 1.5 cloth 高于其自行生产会得到的 1 cloth; A 付出 1.5 cloth 少于自行生产 computer 的 2 cloth, 双方可获益。

### 逐题精讲与评价拓展 | Exam workshop

#### 进一步拆解: 这一课真正考什么

Specialisation 可发生在 worker、firm、region 或 country。Comparative advantage 是国家专业化的机会成本依据; absolute advantage 只比较产量。先算单位机会成本, 再判断贸易方向。

#### 完整因果链 | Cause-and-effect chain

specialisation → repeated practice and capital designed for a task → productivity rises → unit cost falls → output/trade may increase; 同时 dependence 和 structural unemployment risk 可能上升。

写分析题时把每个箭头改成 because 或 therefore 连接的完整句子; 如果中间一步无法解释, 说明链条仍有跳跃。

#### 数字与情境示范 | Numerical or contextual example

A: 8 cloth或4 computers, 1 computer=2 cloth; B:6 cloth或6 computers, 1 computer=1 cloth。B在computer有比较优势, A在cloth有比较优势。

#### English guided explanation

Comparative advantage explains potential gains from trade because resources are allocated to lower-opportunity-cost production. The model assumes transferable resources, acceptable transport costs and mutually beneficial terms of trade.

## Cambridge-style multiple-choice application

Question. A country is less productive than another in both goods. Can it still have comparative advantage?

Answer and reasoning. Yes. It can have a lower opportunity cost in one good even without an absolute advantage. 选择题不仅要找到正确项，还要说明其余方向为什么与定义、公式或因果关系不符。

## Cambridge-style structured application

Question. Discuss whether greater specialisation benefits workers and consumers.

Model development. Productivity and scale may lower prices and raise wages or choice. Workers gain expertise. However, repetitive work can reduce motivation, and demand changes may cause severe unemployment in specialised regions. Consumers become more dependent on supply chains.

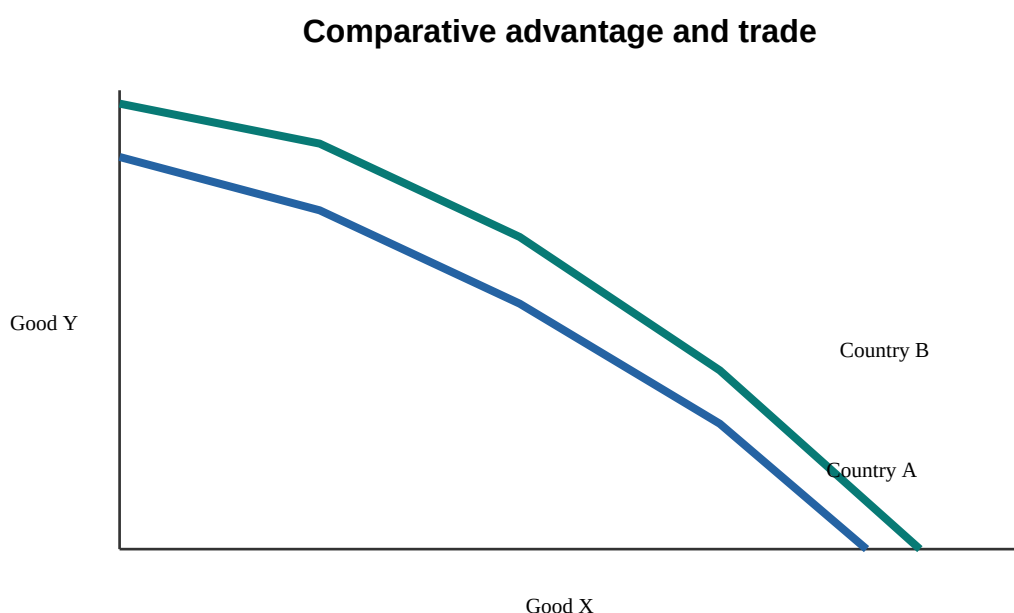
## 高频混淆点 | Common misconception

不要用“谁产得多”判断 comparative advantage; 必须比较 opportunity cost。

## 条件与评价 | Conditions and evaluation

现实收益会被 transport cost、trade barriers、exchange rates、resource mobility 与 adjustment costs 改变。评价不是另起一个无关观点，而是改变时间、幅度、弹性、资源条件或利益相关者后，检查原结论是否仍成立。

## 图表推演 | Diagram analysis



考试图表必须标出坐标轴、曲线名称、新旧均衡和变化方向。文字解释需说明原因、曲线移动以及价格或数量结果。

## 英文精讲 | English explanation

Comparative advantage is determined by lower opportunity cost, not simply higher output.

## IGCSE 大考口径 | Cambridge exam focus

大考口径：准确定义；紧扣题干应用；解释完整因果链；评价结论说明条件。

## 第 30 课 | 2.2 自由贸易与保护主义

所属大章: 国际贸易与全球化 | International trade and globalisation

### 知识点精讲 | Core knowledge

自由贸易可降低价格、扩大选择、形成规模经济并增强竞争。保护主义包括 tariff、quota、subsidy 和 regulations；理由可为幼稚产业、就业、国家安全或反倾销，但代价通常是消费者价格更高、报复与低效率。高分答案必须同时写受保护企业和消费者，而非只写“有利/不利”。

### 深度展开与真题化应用 | In-depth development

#### 自由贸易的收益与调整成本

Free trade 扩大 market size，促进 specialisation、competition、choice 和 economies of scale，并降低进口价格。然而进口竞争会使缺乏竞争力的企业缩小，造成 structural unemployment；国家也可能依赖关键进口或承受外部冲击。

#### 保护工具与利益相关者

Tariff 是进口税，提高进口价格并产生政府收入；quota 直接限制数量，可能产生配额租；subsidy 降低国内企业成本；embargo 禁止贸易。保护可支持 infant industry、employment、strategic goods 和 current account，却提高消费者价格、削弱竞争，并可能引发 retaliation。

#### English development

Protection redistributes income as well as changing total efficiency. Domestic producers and some workers may gain, while consumers and firms using imported inputs may lose. The long-run result depends on whether protected firms become competitive.

#### 真题化知识应用

讨论 steel tariff 时加入 downstream firms: 钢价上涨 → car makers' costs rise → car prices rise/export competitiveness falls。结论比较短期就业保护与长期效率，并说明临时、递减且有绩效条件的 infant-industry support 可能比永久高关税合理。

### 逐题精讲与评价拓展 | Exam workshop

#### 进一步拆解: 这一课真正考什么

自由贸易分析 efficiency、choice、prices、scale 与 export opportunity；保护主义分析 tariff、quota、subsidy、embargo 对 consumers、domestic producers、government 和 downstream firms 的分配影响。

#### 完整因果链 | Cause-and-effect chain

tariff on imported steel → import price rises → demand shifts towards domestic steel → local output/jobs may rise → car makers pay more → car prices rise and export competitiveness may fall → retaliation risk。写分析题时把每个箭头改成 because 或 therefore 连接的完整句子；如果中间一步无法解释，说明链条仍有跳跃。

#### 数字与情境示范 | Numerical or contextual example

世界价\$100, tariff

20%，进口价约\$120。若进口量从1m降至0.7m，理论关税收入约\$14m；但忽略行政成本和可能的逃税。

#### English guided explanation

Protection can support infant industries or strategic capacity, but it reduces competitive pressure and creates costs for users of imported inputs. Quotas restrict quantity directly; tariffs change price and raise government revenue.

### Cambridge-style multiple-choice application

Question. Which group is most likely to lose directly from a tariff on imported food?

Answer and reasoning. Domestic consumers, because they face higher prices and less choice, although local producers may gain. 选择题不仅要找到正确项，还要说明其余方向为什么与定义、公式或因果关系不符。

### Cambridge-style structured application

Question. Discuss whether an infant industry should receive protection.

Model development. Temporary protection may allow learning, scale economies and future competitiveness. It can preserve jobs and diversify the economy. But firms may become dependent and inefficient; consumers pay more and governments may be lobbied to extend protection. Conditions and a credible time limit matter.

### 高频混淆点 | Common misconception

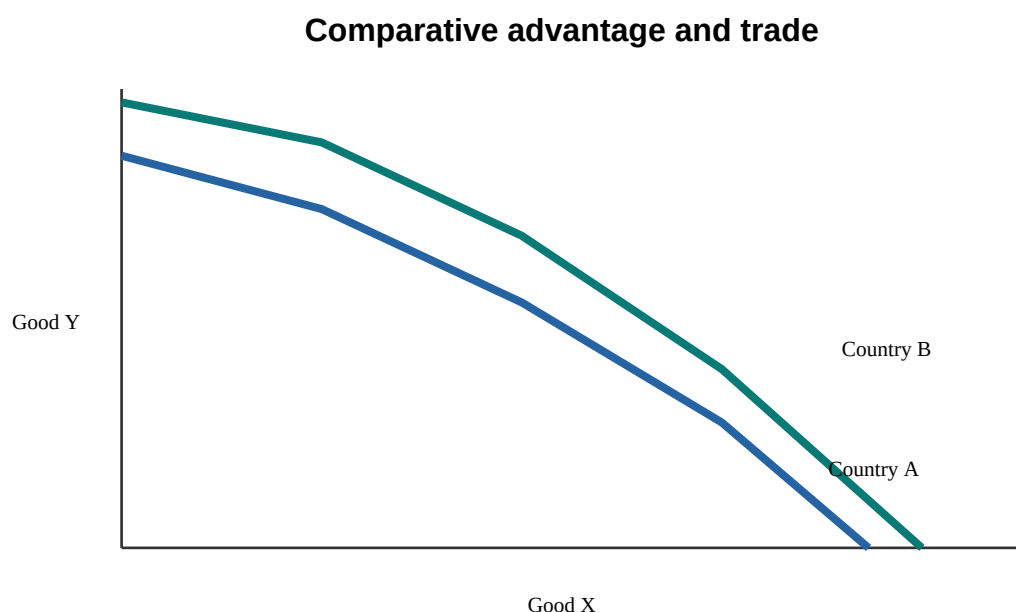
Tariff 不会机械地把 import quantity 降为零；结果取决于需求、国内供给与税率。

### 条件与评价 | Conditions and evaluation

结论要比较短期 adjustment benefits 与长期 efficiency costs，并考虑 retaliation。

评价不是另起一个无关观点，而是改变时间、幅度、弹性、资源条件或利益相关者后，检查原结论是否仍成立。

### 图表推演 | Diagram analysis



考试图表必须标出坐标轴、曲线名称、新旧均衡和变化方向。文字解释需说明原因、曲线移动以及价格或数量结果。

### 英文精讲 | English explanation

Protection may support infant industries, but tariffs and quotas raise prices and reduce competition.

### IGCSE 大考口径 | Cambridge exam focus

大考口径：准确定义；紧扣题干应用；解释完整因果链；评价结论说明条件。

## 第 31 课 | 2.3 汇率与经常账户

所属大章: 国际贸易与全球化 | International trade and globalisation

### 知识点精讲 | Core knowledge

汇率是一个货币用另一个货币表示的价格。货币 depreciation (贬值) 使出口在外国买家眼中更便宜、进口在国内更贵; 若需求有弹性, 净出口可能增加。但进口原料更贵也会推高成本型通胀, 而且短期合约会形成 J-curve。不要写“贬值一定改善贸易差额”。

### 深度展开与真题化应用 | In-depth development

汇率是两种货币的价格

Appreciation 表示本币能兑换更多外币, 使 exports 在国外更贵、imports 对本国更便宜; depreciation 相反。浮动汇率受货币 demand 与 supply 影响: export demand、capital inflow 和高利率可能增加本币需求; import spending 与 capital outflow 增加本币供给。

数量反应决定最终结果

贬值首先改变价格, 但 current account 是否改善取决于 PED、替代品、质量、capacity 和 time lag。进口原材料昂贵会造成 cost-push inflation, 抵消出口价格优势。短期合同固定时数量反应慢, 可能出现 J-curve。

English development

Do not say that appreciation “increases imports” without a mechanism. It lowers the domestic-currency price of imports, which may increase quantity demanded. At the same time, exporters may receive less revenue or lose foreign demand.

真题化知识应用

选择题常考方向: 本币升值受益者可能是进口商和出国游客, 受损者可能是出口企业。讨论贬值时需同时写 exports、imports、inflation 与 debt, 并以需求弹性和进口依赖作判断。

### 逐题精讲与评价拓展 | Exam workshop

进一步拆解: 这一课真正考什么

汇率题先确定“本币能换多少外币”的报价方向, 再判断 appreciation/depreciation。随后分别分析 export price、import price、quantity response、cost-push inflation 和 current account。

完整因果链 | Cause-and-effect chain

currency depreciates → exports cheaper to foreigners/imports dearer domestically → X quantity may rise and M quantity fall → net exports and employment may rise; imported inputs become dearer → costs/inflation may rise。写分析题时把每个箭头改成 because 或 therefore 连接的完整句子; 如果中间一步无法解释, 说明链条仍有跳跃。

数字与情境示范 | Numerical or contextual example

£1=\$1.40 变为 £1=\$1.20, 英镑

depreciation。售价 £100 的商品由 \$140 降至 \$120 (若英镑价不变); 英国买 \$120 商品由约 £85.71 升至 £100。

English guided explanation

An exchange-rate change first alters relative prices. The current account changes only if buyers respond. Elasticity, contracts, production capacity and imported inputs determine the size and timing of the quantity

response.

### Cambridge-style multiple-choice application

Question. Who is most likely to gain from an appreciation: an exporter paid in foreign currency, a domestic tourist travelling abroad, or a producer competing with cheaper imports?

Answer and reasoning. The tourist travelling abroad, because each unit of domestic currency buys more foreign currency. 选择题不仅要找到正确项，还要说明其余方向为什么与定义、公式或因果关系不符。

### Cambridge-style structured application

Question. Discuss whether depreciation will reduce a current-account deficit.

Model development. Cheaper exports and dearer imports may raise export receipts and reduce import spending. Yet demand may be inelastic, firms may lack capacity, and imported raw materials raise costs. Short-run contracts can delay adjustment. Success depends on elasticities and time.

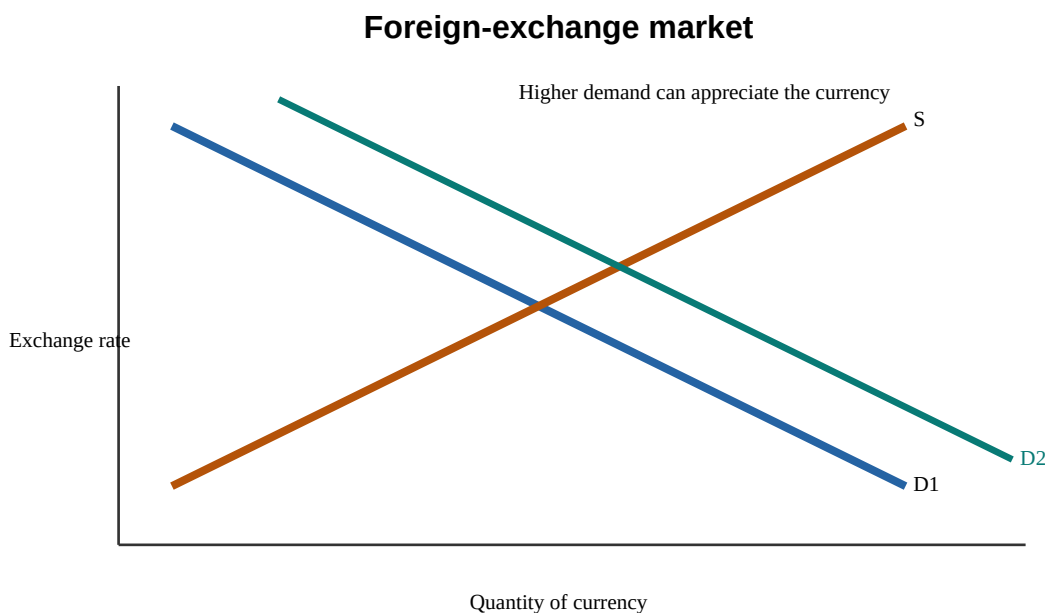
### 高频混淆点 | Common misconception

Appreciation 不是“出口数量必然下降”；先是相对价格改变，再由 PED 决定数量。

### 条件与评价 | Conditions and evaluation

同时考虑 consumers、exporters、importers、inflation、foreign-currency debt 和 central-bank objectives。评价不是另起一个无关观点，而是改变时间、幅度、弹性、资源条件或利益相关者后，检查原结论是否仍成立。

## 图表推演 | Diagram analysis



考试图表必须标出坐标轴、曲线名称、新旧均衡和变化方向。文字解释需说明原因、曲线移动以及价格或数量结果。

## 英文精讲 | English explanation

Appreciation makes exports dearer and imports cheaper; depreciation has the opposite price effect.

## IGCSE 大考口径 | Cambridge exam focus

大考口径：准确定义；紧扣题干应用；解释完整因果链；评价结论说明条件。



## 第 32 课 | 2.4 全球化

所属大章: 国际贸易与全球化 | International trade and globalisation

### 知识点精讲 | Core knowledge

全球化提高资本、信息、商品和人员流动，可能带来 FDI、就业和技术转移，也可能令弱劳动者面临跨国竞争、利润外流和文化/环境压力。判断题应结合一个国家的技能、制度与产业结构。

### 深度展开与真题化应用 | In-depth development

#### 全球化不只是贸易增加

Globalisation 指商品、服务、资本、技术、信息和生产跨国联系增强。Multinational companies 通过 FDI 建厂，可能带来 capital、employment、training、tax revenue 和 access to export markets；东道国也可能面对 profit repatriation、tax avoidance、environmental damage 和对本地企业的竞争。

#### 不同群体获得不同结果

消费者可能因进口和竞争获得更低价格与更多选择；出口企业进入更大市场；部分工人获得岗位，另一些工人因 offshoring 失业。来源国企业可降低成本，却可能减少本国某些就业。评价必须分别讨论 consumers、workers、firms、government 和 environment。

#### English development

Globalisation can raise productivity through technology transfer and competition, but benefits are not automatic or equally distributed. Regulation, labour skills, infrastructure and bargaining power affect how much value remains in the host economy.

#### 真题化知识应用

讨论 MNC 是否提高 developing country living standards: 正面写 jobs → income → consumption、training → productivity、tax → public services；反面写 low wages、profits leave、pollution、local firms close。结论取决于 regulation、local linkages、tax agreements 和 job quality。

### 逐题精讲与评价拓展 | Exam workshop

#### 进一步拆解：这一课真正考什么

全球化包括 trade、FDI、finance、technology、information 与 labour links。MNC 是重要载体，但全球化不等于只有跨国公司。按 home country、host country 与 stakeholder 分析收益成本。

#### 完整因果链 | Cause-and-effect chain

MNC builds factory → capital and jobs enter host economy → workers receive income/training → local suppliers gain orders → tax revenue may fund services；若 inputs imported and profits repatriated, local multiplier 较小。写分析题时把每个箭头改成 because 或 therefore 连接的完整句子；如果中间一步无法解释，说明链条仍有跳跃。

#### 数字与情境示范 | Numerical or contextual example

工厂销售\$100m，但进口投入\$45m、利润汇回\$25m，本地工资税收采购只有\$30m。Gross sales 不能代表留在东道国的全部收益。

#### English guided explanation

Technology transfer and access to global markets can raise productivity. However, bargaining power, tax rules, environmental regulation and local linkages determine how gains are shared. Globalisation creates

winners and losers within countries.

### Cambridge-style multiple-choice application

Question. Which is a likely host-country benefit of FDI: guaranteed equal income, technology transfer, no competition, or elimination of imports?

Answer and reasoning. Technology transfer, although its extent depends on training and links with domestic firms. 选择题不仅要找到正确项，还要说明其余方向为什么与定义、公式或因果关系不符。

### Cambridge-style structured application

Question. Discuss whether MNCs improve living standards in developing economies.

Model development. They may create employment, wages, training, exports and tax revenue. Competition and technology can raise productivity. Conversely, profits may leave, local firms may close, jobs may be insecure and pollution may rise. Regulation and job quality determine the net effect.

### 高频混淆点 | Common misconception

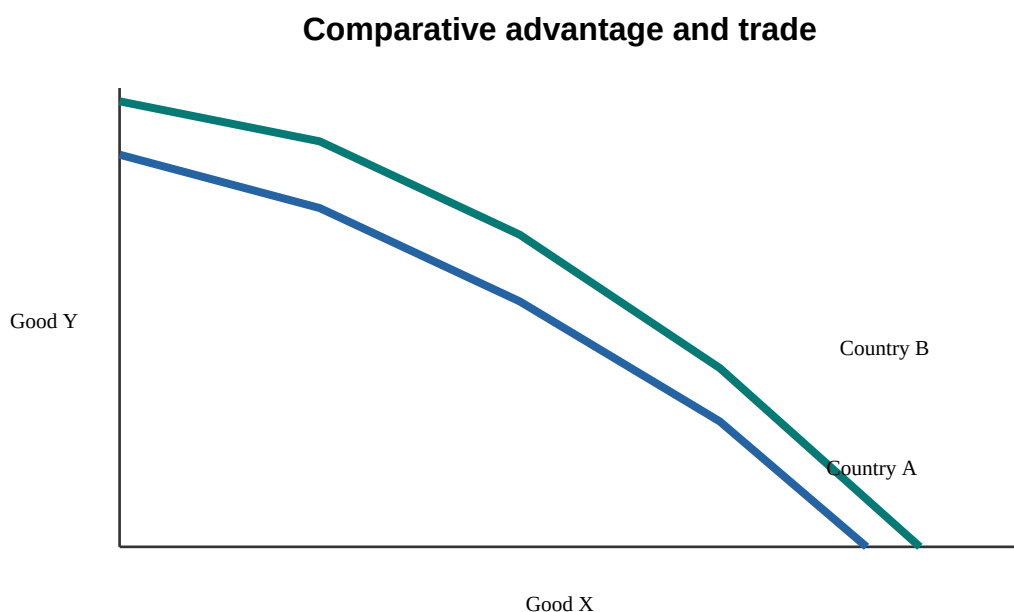
Export growth不等于所有居民收入提高；需追踪 ownership、wages、tax 和 distribution。

### 条件与评价 | Conditions and evaluation

短期 job creation 与长期 capability building 要分开，尤其关注技能转移和本地供应链。

评价不是另起一个无关观点，而是改变时间、幅度、弹性、资源条件或利益相关者后，检查原结论是否仍成立。

## 图表推演 | Diagram analysis



考试图表必须标出坐标轴、曲线名称、新旧均衡和变化方向。文字解释需说明原因、曲线移动以及价格或数量结果。

### 英文精讲 | English explanation

Globalisation increases international flows of goods, capital and production, but gains and costs are uneven.

### IGCSE 大考口径 | Cambridge exam focus

大考口径：准确定义；紧扣题干应用；解释完整因果链；评价结论说明条件。

## 第 33 课 | 2.5 比较优势的数值法

所属大章：国际贸易与全球化 | International trade and globalisation

### 知识点精讲 | Core knowledge

若 A 国一小时产 8 件衣服或 4 台电脑，生产 1 台电脑的机会成本是 2 件衣服；B 国一小时产 6 件衣服或 6 台电脑，生产 1 台电脑的机会成本是 1 件衣服。因此 B 在电脑有比较优势，A 在衣服有比较优势。比较的是比率，不能只看“谁生产得更多”。

汇率题先说明报价方式。若 1 USD 从 7.0 CNY 升至 7.3

CNY，人民币对美元贬值；中国出口对美国买家更便宜，但进口原材料更贵。紧接着加一句条件：The effect on the current account depends on price elasticities and the time needed for contracts to change.

### 深度展开与真题化应用 | In-depth development

先统一“每一单位”的方向

若表格给的是相同资源下最大产量，用“另一商品产量 ÷ 本商品产量”求一单位机会成本。A 可产 12 wheat 或 6 cars，则 1 car=2 wheat，1 wheat=0.5 car。每个国家都算两行，再横向比较同一商品；机会成本较低者有 comparative advantage。

检验交换比率是否互利

Terms of trade 必须介于两国国内机会成本之间。若 A 生产 1 car 放弃 2 wheat，B 放弃 4 wheat，国际交换率为 1 car:3 wheat，则 A 出口 car 可换 3 wheat，超过自行生产得到的 2；B 进口 car 只付 3 wheat，少于自行生产放弃的 4。

English development

Write the unit beside every opportunity-cost answer. A number without “units of wheat per car” is easy to reverse. Comparative advantage is identified product by product, and mutually beneficial trade also requires an acceptable exchange ratio.

真题化知识应用

常见错误是比较 12 与 8 后直接判优势，或把分子分母颠倒。检查方法：一个国家两种机会成本的乘积应为 1；若不是，通常计算方向有误。

### 逐题精讲与评价拓展 | Exam workshop

进一步拆解：这一课真正考什么

任何数值表先写“生产1单位X放弃多少Y”。相同资源下最大产量题用 other output/target output。每国计算两个方向，乘积应接近1，可作为反向检查。

完整因果链 | Cause-and-effect chain

identify lower opportunity cost → specialise partly or fully → total combined output may rise → choose terms of trade between domestic costs → both countries can consume beyond pre-trade possibilities。写分析题时把每个箭头改成 because 或 therefore 连接的完整句子；如果中间一步无法解释，说明链条仍有跳跃。

数字与情境示范 | Numerical or contextual example

A:12 wheat或6 cars，所以1 car=2 wheat；B:8 wheat或2 cars，所以1 car=4 wheat。A出口car。若1 car换3 wheat，A得到超过2，B支付少于4，双方获益。

English guided explanation

A mutually beneficial exchange ratio lies strictly between the two domestic opportunity costs. Specialisation need not be complete when transport costs, risk, increasing opportunity cost or strategic concerns are included.

### Cambridge-style multiple-choice application

Question. Country A gives up 2 rice for one phone; B gives up 5. Which exchange rate benefits both?

Answer and reasoning. One phone for any amount strictly between 2 and 5 rice, such as 3 rice.

选择题不仅要找到正确项，还要说明其余方向为什么与定义、公式或因果关系不符。

### Cambridge-style structured application

Question. Using opportunity-cost data, explain why two countries may gain from trade.

Model development. The country with the lower opportunity cost should specialise relatively more in that product. Combined output can rise because resources are used where sacrifice is smaller. At terms of trade between the two costs, the exporter receives more than domestically and the importer gives up less.

### 高频混淆点 | Common misconception

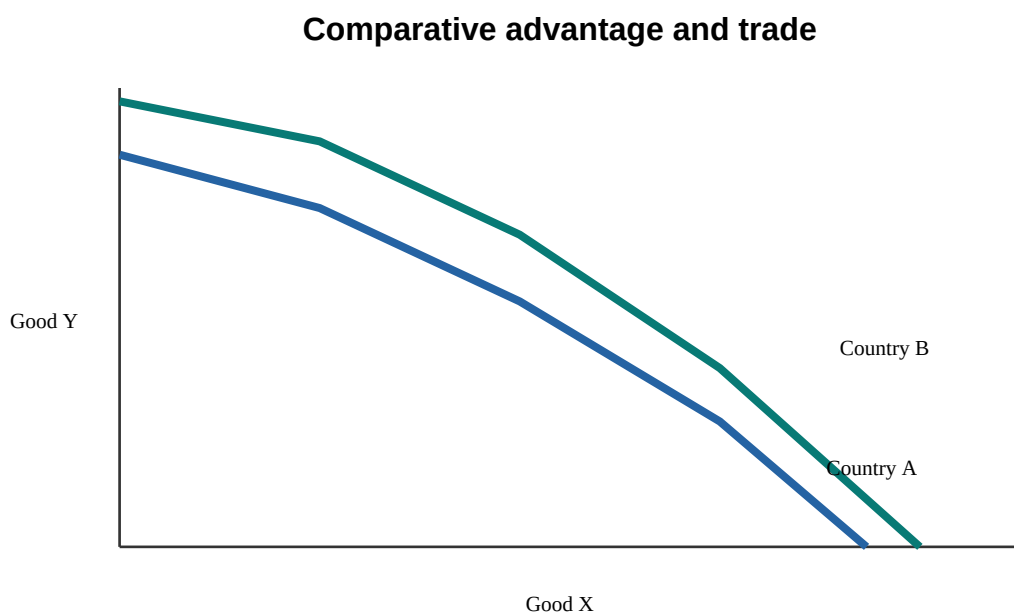
分子分母颠倒会同时错判两个产品。答案必须写单位，例如 wheat per car。

### 条件与评价 | Conditions and evaluation

模型结论还取决于交换比率、运输成本、需求、资源能否转移和贸易收益如何分配。

评价不是另起一个无关观点，而是改变时间、幅度、弹性、资源条件或利益相关者后，检查原结论是否仍成立。

## 图表推演 | Diagram analysis



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### 英文精讲 | English explanation

Define the key term precisely, apply it to the case material and develop a complete economic explanation.

### IGCSE 大考口径 | Cambridge exam focus

大考口径：准确定义；紧扣题干应用；解释完整因果链；评价结论说明条件。